# Pseudorandomness of UFLM: A Characterization via Its Linear Layer

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**Abstract**

This paper systematically analyzes the security of the two-branch Unified Feistel Lai Massey (UFLM) structure with independent random round functions under chosen plaintext and chosen ciphertext attacks, focusing on its indistinguishability from a random permutation. UFLM uses an invertible linear layer represented as a $\boldsymbol{2 \times 2}$ block matrix $\boldsymbol{\varphi}$ with blocks $\boldsymbol{A_{11}, A_{12}, A_{21}, A_{22}}$. Previously, Dai et al. proved that when $\boldsymbol{A_{12}}$ is invertible, $\boldsymbol{4}$-round UFLM achieves CCA security and resists up to $\boldsymbol{\mathcal{O}(2^{n/2})}$ queries, where the UFLM input is $\boldsymbol{2n}$ bits. Our work imposes no restriction on $\boldsymbol{A_{12}}$. We determine the minimal number of rounds for UFLM to achieve CPA and CCA security, fully determined by the parameters $\boldsymbol{T(A_{12}^{\top}, A_{11}^{\top})}$ and $\boldsymbol{T(A_{12}, A_{22})}$. For UFLM with enough rounds to be secure, the query bound is primarily determined by the rank of $\boldsymbol{A_{12}}$. For all UFLM with too few rounds to be secure, we present successful distinguishing attacks that require at most four queries. Our results rigorously show, for the first time, that when $\boldsymbol{A_{12}}$ has full rank, UFLM requires the fewest rounds to achieve CPA and CCA security and attains the highest query bound. Nevertheless, when $\boldsymbol{A_{12}}$ is not full rank, CPA and CCA security can still be achieved by increasing the number of rounds unless $\boldsymbol{T(A_{12}^{\top}, A_{11}^{\top}) = \infty}$ or $\boldsymbol{T(A_{12}, A_{22}) = \infty}$. At last, for involutory $\boldsymbol{\varphi}$, we find UFLM achieves CPA and CCA security if and only if $\boldsymbol{A_{12}}$ has full rank.

**Keywords:** UFLM, Feistel, Lai-Massey, Distinguishing attack.

## 1 Introduction

Modern block ciphers are predominantly constructed from iterative round-based structures that combine nonlinear functions with linear diffusion layers. Three major paradigms have shaped the development of block cipher designs: Feistel structures [1], substitution–permutation networks (SPN) [2], and the Lai-Massey structure [3]. Among these, Feistel and Lai-Massey stand out as two-branch designs that support *inverse-free decryption*, meaning decryption does not require inverting the round function. This property has enabled efficient implementations and has been adopted in classic and modern ciphers such as DES [4], CAST [5], SIMON [6], Simeck [7], IDEA [8, 9], FOX [10], and Ballet [11].

**Pseudorandomness.** The security of a cryptographic structure is measured by its pseudorandomness, that is, indistinguishability from an ideal random permutation under standard attack models. These security notions are formally defined as pseudo-random permutation (PRP) security against chosen-plaintext attacks (CPA) and strong pseudo-random permutation (sPRP) security against chosen-ciphertext attacks (CCA). A structure that fails to satisfy this property is vulnerable to distinguishing attacks that tell its construction apart from an ideal random

**Feistel, Lai-Massey, and UFLM.** Although Feistel and Lai-Massey appear different, decades of analysis have shown that their security behaviors, especially pseudorandomness, are strikingly similar. Classical results [1, 3, 12, 13] demonstrate that:

- 2-round instantiations are CPA-insecure with only two queries;
- 3-round variants achieve PRP but not sPRP security;
- 4-round variants achieve full sPRP security up to $\mathcal{O}(2^{n/2})$ queries.

These parallel results suggest that both structures obey a deeper unifying principle. Previous attempts at unification include quasi-Feistel systems [14] and Feistel-like generalizations [15, 16], but these frameworks either do not encompass Lai-Massey fully or depend on restrictive assumptions on the linear transformation.

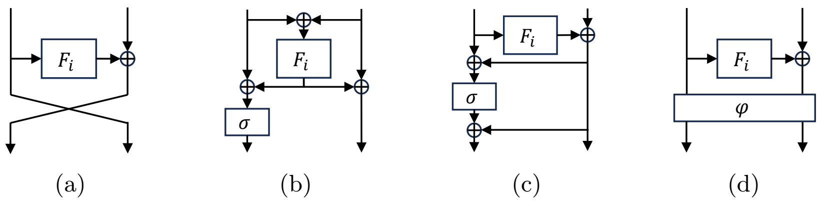



**Fig. 1**: Round transformations of Feistel, Lai-Massey, the affine equivalent variant of Lai-Massey, and UFLM.

In 2024, Dai et al. [17] introduced the *Unified-Feistel-Lai-Massey (UFLM)* framework. UFLM captures Feistel and the affine equivalent variant of Lai-Massey within a single algebraic model by identifying that their only structural difference lies in the choice of a $2n \times 2n$ invertible linear layer

$$\varphi = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}. \tag{1}$$

Figs. 1a to 1d illustrate this relation. The framework preserves the inverse-free property and offers a systematic way to reason about two-branch ciphers, opening the door to new constructions and more unified analyses.

In each round of UFLM structure, the round function $F_i$ is applied to one half, and then the output is XORed with the other half. Finally, a public invertible linear mapping $\varphi$ is applied to both halves, where $A_{ij}$ for $i, j \in \{1, 2\}$ are matrices over $n$ bits. This final mapping $\varphi$ is usually omitted in the last round. They investigated the number of rounds of impossible differentials, zero correlation linear hulls, and integral distinguishers of UFLM instances with bijective round functions based on different orders of the linear transformation adopted. Their analysis insightfully attributes the known superior security of Lai-Massey against certain cryptanalytic attacks (e.g., impossible differentials) to the higher algebraic order of its linear map $\varphi$ compared to the simple swap (order 2) used in a Feistel network.

**Motivations.** Although the UFLM framework unifies Feistel and the affine equivalent variant of Lai-Massey, its indistinguishability properties remain only partially understood. The only known result is that 4-round UFLM achieves sPRP security when the submatrix $A_{12}$ is invertible [17]. In the Feistel structure of Fig. 1a the matrix $A_{12}$ is the identity $I_n$, while in the Lai–Massey structure of Fig. 1c it equals $\sigma \oplus I_n$ for an orthomorphism $\sigma$, and both matrices are invertible.

This assumption, however, is restrictive and excludes many linear layers that may naturally arise in practical designs or be preferred for implementation reasons. When $A_{12}$ is non-invertible, empirical evidence indicates that internal collisions may occur, potentially enabling low-query distinguishing attacks. Yet there is currently no systematic theory explaining how $A_{12}$ influences randomness propagation, how non-invertibility leads to deterministic differentials, and how many rounds are needed to recover PRP or sPRP security.

A closer inspection of the UFLM round transformation shows that $A_{12}$ plays a central role in transferring randomness from the branch processed by the round function to the other branch. Before applying $\varphi$, only the right branch contains fresh randomness from $F_i$; after applying $\varphi$, the left branch becomes a linear combination involving $A_{12}$, which then feeds into the next round. Thus, the algebraic structure of $A_{12}$ determines whether randomness propagates effectively or collapses. If $A_{12}$ is invertible, the diffusion resembles that in Feistel and Lai-Massey. If it is non-invertible, many-to-one mappings and deterministic images may arise, creating exploitable patterns.

These observations lead to the central question of this work: *To what extent can the pseudorandomness of UFLM be explained and fully characterized through the algebraic* *structure of its linear layer, in particular the influence of the submatrix $A_{12}$, under both CPA and CCA security models?*

**Our contributions.** We systematically evaluate the security of UFLM and establish how its linear layer, most notably the block $A_{12}$, dictates the structure's pseudorandomness. For matrices $n \times n$ matrices $X$ and $Y$ over $GF(2)$ and a integer $i$, let

$$V_{X,Y,i} \coloneqq \begin{pmatrix} X \\ XY \\ \vdots \\ XY^{i-1} \\ XY^{i} \end{pmatrix}, \tag{2}$$

$$T(X,Y) \coloneqq \min\{\ i \mid \mathrm{rank}(V_{X,Y,i}) = n\}, \tag{3}$$

which implies $i = T(X,Y)$ is the smallest index such that $V_{X,Y,i}$ achieves full rank. Our analysis reveals that the secure round threshold for UFLM can be characterized by the quantities $T(A_{12}, A_{22})$ and $T(A_{12}^{\top}, A_{11}^{\top})$. Once the secure round threshold is reached, the security strength of UFLM is primarily governed by $\mathrm{rank}(A_{12})$. Let $r$ be the number of UFLM rounds. We summarize our results as follows.

- **When $T(A_{12}, A_{22}) < \infty$ and $T(A_{12}^{\top}, A_{11}^{\top}) < \infty$:**
  - **For $1 \le r \le T(A_{12}, A_{22}) + T(A_{12}^{\top}, A_{11}^{\top}) + 2$,** UFLM is CPA-insecure with at most two queries.
  - **For $r = T(A_{12}, A_{22}) + T(A_{12}^{\top}, A_{11}^{\top}) + 3$,** UFLM achieves PRP security up to $\mathcal{O}(\frac{1}{r} \cdot 2^{\frac{\mathrm{rank}(A_{12})}{2}})$ queries.
  - **For $r < \max\{T(A_{12}, A_{22}) + T(A_{12}^{\top}, A_{11}^{\top}) + 3, 2T(A_{12}, A_{22}) + 4\}$,** UFLM is CCA-insecure with at most four queries, except that the case $r = 2T(A_{12}, A_{22}) + 3$ requires an additional condition to allow the attack.
  - **For $r = \max\{T(A_{12}, A_{22}) + T(A_{12}^{\top}, A_{11}^{\top}) + 3, 2T(A_{12}, A_{22}) + 4\}$,** UFLM achieves sPRP security up to $\mathcal{O}(\frac{1}{r} \cdot 2^{\frac{\mathrm{rank}(A_{12})}{2}})$ queries.
- **When $T(A_{12}, A_{22}) = \infty$,** UFLM is CPA-insecure for any number of rounds with only two queries, regardless of the number of rounds.
- **When $T(A_{12}^{\top}, A_{11}^{\top}) = \infty$,** UFLM is CPA-insecure for any number of rounds with merely one query, regardless of the number of rounds.

These results show that whenever $T(A_{12}, A_{22}) < \infty$ and $T(A_{12}^{\top}, A_{11}^{\top}) < \infty$, UFLM can always achieve both CPA and CCA security with a sufficiently large number of rounds, irrespective of whether $A_{12}$ is invertible.

**When $\mathrm{rank}(A_{12}) = n$,** it follows that $T(A_{12}, A_{22}) = T(A_{12}^{\top}, A_{11}^{\top}) = 0$. Consequently, applying the general results to this special case shows that the security of UFLM matches that of Feistel and Lai–Massey structures:

- **For $1 \le r \le 2$,** UFLM is CPA-insecure with at most two queries.
- **For $r = 3$,** UFLM achieves PRP security up to $\mathcal{O}(2^{\frac{n}{2}})$ queries.
- **For $1 \le r \le 3$,** UFLM is is CCA-insecure with at most three queries.
- **For $r = 4$,** UFLM achieves full sPRP security up to $\mathcal{O}(2^{\frac{n}{2}})$ queries.

In this scenario, UFLM attains both CPA and CCA security with the minimal number of rounds, achieving the highest security level.

An involutory matrix with $\varphi^{-1} = \varphi$ allows UFLM to achieve encryption-decryption consistency, greatly reducing the hardware footprint and making it a valuable design component. When UFLM employs such a special linear layer, its security is characterized as follows.

- **When** $\mathbf{rank}(\boldsymbol{A_{12}}) = \boldsymbol{n}$, the security of UFLM coincides with the standard case.
- **When** $\mathbf{rank}(\boldsymbol{A_{12}}) < \boldsymbol{n}$, we have $T(A_{12}, A_{22}) = T(A_{12}^\top, A_{11}^\top) = \infty$. Hence, UFLM is CPA-insecure with merely one query for any number of rounds.

This implies that for UFLM to achieve security when an involutory matrix $\varphi$ is employed as its linear layer, the submatrix $A_{12}$ must be invertible.

# 2 Preliminaries

## 2.1 Notations

We write $\{0,1\}^n$ for the set of all binary strings of length $n$, $\mathsf{Perm}(n)$ for the family of permutations over $\{0,1\}^n$, and $\mathsf{Func}(n,m)$ for the family of functions mapping $\{0,1\}^n$ to $\{0,1\}^m$. For a set $\mathcal{X}$, let $|\mathcal{X}|$ be the number of all elements in $\mathcal{X}$. For a binary string $x$, we denote its left half by $x_{|L}$ and its right half by $x_{|R}$. For a function $f$, we write $f_{|L}$ for the function that returns the left half of $f$'s output, and $f_{|R}$ for the function that returns the right half. The notation $\$^n$ represents a function returning a uniformly random $n$-bit string on each call. An adversary $\mathcal{A}$ with oracle access to $f$ that outputs a bit $b$ is denoted by $\mathcal{A}^f \Rightarrow b$. Let $x := y$ indicates defining $x$ as $y$. The notation $x \leftarrow\!\!\$\ \mathcal{X}$ means that $x$ is chosen uniformly at random from the set $\mathcal{X}$. For binary vectors $x = x_0x_1 \dots x_{n-1}$ and $y = y_0y_1 \dots y_{n-1}$ with $x_i, y_i \in \{0,1\}$, we use $x \oplus y$ to denote bitwise exclusive-OR and $x \cdot y := x_0y_0 \oplus x_1y_1 \oplus \cdots \oplus x_{n-1}y_{n-1}$ to denote their inner product over $GF(2)$. The symbol $C(n,m)$ denotes the binomial coefficient, defined as $\frac{n!}{m!(n-m)!}$, which counts the combinations of $m$ elements chosen from $n$ distinct elements. For a $n$-bit binary string $x = x_0x_1 \dots x_{n-1}$, it can be interpreted in context as an $n$-dimensional column vector $(x_0, x_1, \dots, x_{n-1})^\top$ in $GF(2)$. Let $O_{n\times m}$ denote the $n \times m$ zero matrices. Let $I_n$ and $O_n$ denote the $n \times n$ identity and zero matrices, respectively.

## 2.2 Security Definitions

**(Strong) Pseudorandom Permutation, (s)PRP** Let $\mathbf{E} : \mathcal{K} \times \{0,1\}^n \to \{0,1\}^n$ be a keyed permutation, $\pi \leftarrow\!\!\$\ \mathrm{Perm}(n)$ be a random permutation. Let adversary $\mathcal{A}$ be allowed to query the forward oracle. The PRP advantage of $\mathcal{A}$ is defined as: $\mathrm{Adv}_{\mathbf{E}}^{\mathrm{prp}}(\mathcal{A}) := |\Pr_{K\leftarrow\$\mathcal{K}}[\mathcal{A}^{\mathbf{E}_K} \Rightarrow 1] - \Pr_{\pi\leftarrow\$\mathrm{Perm}(n)}[\mathcal{A}^{\pi} \Rightarrow 1]|$. Let adversary $\mathcal{A}$ be allowed to query both the forward oracle and the backward oracle. The sPRP advantage of $\mathcal{A}$ is defined as: $\mathrm{Adv}_{\mathbf{E}}^{\mathrm{sprp}}(\mathcal{A}) := |\Pr_{K\leftarrow\$\mathcal{K}}[\mathcal{A}^{\mathbf{E}_K,\mathbf{E}_K^{-1}} \Rightarrow 1] - \Pr_{\pi\leftarrow\$\mathrm{Perm}(n)}[\mathcal{A}^{\pi,\pi^{-1}} \Rightarrow 1]|$.

**Pseudorandom Function, PRF** Let $F : \mathcal{K} \times \{0,1\}^n \to \{0,1\}^m$ be a keyed function, $f \leftarrow\$ \mathrm{Func}(n, m)$ be a random function. The PRF advantage of adversary $\mathcal{A}$ is defined as: $\mathrm{Adv}_F^{\mathrm{prf}}(\mathcal{A}) \coloneqq |\Pr_{K\leftarrow\$\mathcal{K}}[\mathcal{A}^{F_K} \Rightarrow 1] - \Pr_{f\leftarrow\$\mathrm{Func}(n,m)}[\mathcal{A}^f \Rightarrow 1]|$.

## 2.3 UFLM Structure

Dai et al. [17] proposed a new two-branch framework called UFLM, which contains both Feistel and Lai-Massey structures as its instances. Let $F_i$ be a round function on $\{0,1\}^n$ in the $i$-th round, where $F_i$ is keyed by the subkey $K_i$ chosen randomly from key space $\mathcal{K}$. Generally, $F_i$ is nonlinear. Let the public invertible linear mapping $\varphi$ be defined in Eqs. (1) and (4). Let the inverse of the public linear mapping $\varphi$ be

$$\varphi^{-1} = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}. \tag{4}$$

Given an input $(Y_1, Y_0)$, the output of $r$-round UFLM is $(Y_{3r-2}, Y_{3r-1})$, where $Y_{3i-1} = F_i(Y_{3i-2}) \oplus Y_{3i-3}$, for $i = 1, 2, \ldots, r$, and $(Y_{3i+1}, Y_{3i}) = (A_{11}Y_{3i-2} \oplus A_{12}Y_{3i-1}, A_{21}Y_{3i-2} \oplus A_{22}Y_{3i-1})$ for $i = 1, 2, \ldots, r-1$.

The round transformation for the $i$-th UFLM round is defined as $\mathbf{R}_i(X, Y) \coloneqq (A_{11}X \oplus A_{12}(F_i(X) \oplus Y), A_{21}X \oplus A_{22}(F_i(X) \oplus Y))$ (See Fig. 1d) with variant $\mathbf{R}'_i(X, Y) \coloneqq (X, F_i(X) \oplus Y)$ for $X, Y \in \{0,1\}^n$.

The UFLM encryption algorithm from round $i$ to round $j$ is defined as $\mathbf{E}_{[i,j]} \coloneqq \mathbf{R}'_j \circ \mathbf{R}_{j-1} \circ \cdots \circ \mathbf{R}_i$, with the shorthand $\mathbf{E}_{[j]} \coloneqq \mathbf{E}_{[1,j]}$. The variant of the encryption algorithm is defined as $\mathbf{E}'_{[i,j]} = \varphi \circ \mathbf{E}_{[i,j]}$, with the shorthand notation $\mathbf{E}'_{[j]} \coloneqq \mathbf{E}'_{[1,j]}$. The UFLM decryption from round $j$ back to round $i$ is defined as $\mathbf{D}_{[i,j]} \coloneqq \mathbf{E}^{-1}_{[i,j]}$, with the shorthand $\mathbf{D}_{[j]} \coloneqq \mathbf{D}_{[1,j]}$. The variant of the decryption algorithm is defined as $\mathbf{D}'_{[i,j]} = \varphi^{-1} \circ \mathbf{D}_{[i,j]}$, with the shorthand $\mathbf{D}'_{[j]} \coloneqq \mathbf{D}'_{[1,j]}$.

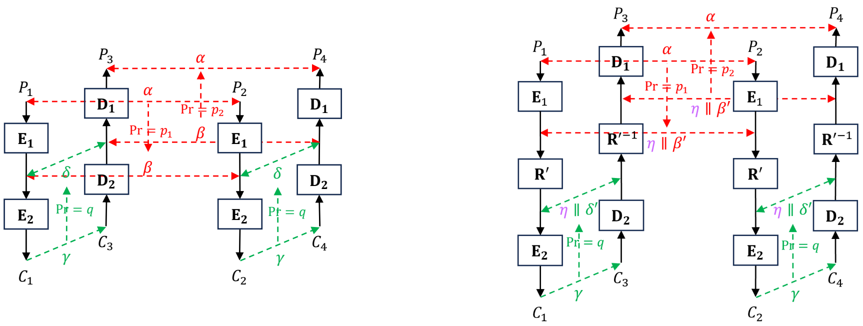
## 2.4 Boomerang Attacks

Boomerang attack [18] is used for distinguishing attack or key recovery attack against encryption algorithm $\mathbf{E}$ in the CCA setting.

**Boomerang attack against $\mathbf{E} \coloneqq \mathbf{E}_2 \circ \mathbf{E}_1$.** A standard Boomerang attack is illustration is provided in Fig. 2a. Divide the encryption algorithm into two parts: $\mathbf{E}_1$ and $\mathbf{E}_2$. Let $\mathbf{D}_1$ and $\mathbf{D}_2$ be the inverse algorithms of $\mathbf{E}_1$ and $\mathbf{E}_2$, respectively. The core of the attack exploits three high-probability differential propagations $\alpha \xrightarrow{\mathbf{E}_1} \beta, \beta \xrightarrow{\mathbf{D}_1} \alpha, \gamma \xrightarrow{\mathbf{D}_2} \delta$ with probabilities $\Pr[\alpha \xrightarrow{\mathbf{E}_1} \beta] \coloneqq p_1, \Pr[\beta \xrightarrow{\mathbf{D}_1} \alpha] \coloneqq p_2, \Pr[\gamma \xrightarrow{\mathbf{D}_2} \delta] \coloneqq q$, respectively. The attack proceeds as follows

(1) Randomly select a pair of plaintexts $(P_1, P_2)$ which satisfies $P_1 \oplus P_2 = \alpha$, query the encryption oracle to obtain the ciphertext pair $(C_1, C_2)$.
(2) Query the decryption oracle for $C_3 = C_1 \oplus \gamma$ and $C_4 = C_2 \oplus \gamma$ to obtain the plaintext pair $(P_3, P_4)$.

Then $(P_3, P_4)$ satisfies $P_3 \oplus P_4 = \alpha$ with probability $p_1 p_2 q^2$. In contrast, for a random permutation, under the same process, $P_3 \oplus P_4 = \alpha$ with probability at most $\frac{1}{2^{n-1}}$ due




(a) (b)

**Fig. 2**: Boomerang attacks. (a) Boomerang attack against $\mathbf{E} \coloneqq \mathbf{E}_2 \circ \mathbf{E}_1$. (b) Boomerang attack against $\mathbf{E} \coloneqq \mathbf{E}_2 \circ \mathbf{R}' \circ \mathbf{E}_1$, where $\mathbf{R}'$ is the round transformation of Feistel without the final swap.

to randomness. This enables a distinguishing attack with success probability at least $p_1 p_2 q^2 - \frac{1}{2^{n-1}}$, using only two encryption and two decryption queries.

**Boomerang attack against $\mathbf{E} \coloneqq \mathbf{E}_2 \circ \mathbf{R}' \circ \mathbf{E}_1$.** In the above boomerang attack, when the left halves of $\beta$ and $\delta$ are identical, inserting the round transformation of Feistel without the final swap, defined as $\mathbf{R}'(X, Y) \coloneqq (X, F(X) \oplus Y)$, such that the encryption becomes $\mathbf{E} = \mathbf{E}_2 \circ \mathbf{R}' \circ \mathbf{E}_1$, still allows the Boomerang attack to succeed with its probability unchanged. This result is based on *Feistel switch* [18]. We write $\beta = \eta \parallel \beta'$ and $\delta = \eta \parallel \delta'$; the attack procedure is shown in Fig. 2b.

# 3 Symmetry and Algebraic Properties of $\varphi$ and $\varphi^{-1}$

This section presents the symmetry between the parameters obtained from the linear layer $\varphi$ of UFLM and those obtained from its inverse $\varphi^{-1}$. Furthermore, we analyze how the rank of $A_{12}$ directly influences these parameters and, as a special case, characterize the rank-stagnation phenomenon when $\varphi$ is involutory. These will significantly simplify our subsequent analysis of UFLM security and provide deeper insight into its indistinguishability security.

## 3.1 Equivalence of Ranks Between $A_{12}$ and $B_{12}$

In subsequent sections, we will demonstrate that when UFLM achieves (s)PRP security, its security strength is related to $\mathrm{rank}(A_{12})$ for encryption, while it is related to $\mathrm{rank}(B_{12})$ for decryption. Hence, this section investigates the relationship between these ranks. Our results will reveal a perfect symmetry between them. The main lemma is stated as follows.

*Lemma 1* For any $\varphi, \varphi^{-1}$ defined in Eqs. (1) and (4), it holds that

$$\mathrm{rank}(A_{12}) = \mathrm{rank}(B_{12}).$$

*Proof* For an $n \times n$ matrix $M$ acting on $n$-bit vectors, let $\ker(M)$ denote the kernel of $M$, defined as $\ker(M) = \{x : Mx = 0^n\}$. By the rank-nullity theorem, for any matrix $M$,

$$\dim(\ker(M)) + \mathrm{rank}(M) = n.$$

Thus, to show $\mathrm{rank}(A_{12}) = \mathrm{rank}(B_{12})$, it suffices to prove $\dim(\ker(A_{12})) = \dim(\ker(B_{12}))$. We demonstrate the following two statements.

(1) $\dim(\ker(B_{12})) \leq \dim(\ker(A_{12}))$: Let $x \in \ker(B_{12})$. From the equation $\varphi\varphi^{-1} = I_{2n}$, we obtain

$$\begin{cases} A_{11}B_{12} \oplus A_{12}B_{22} = O_n, & (5) \\ A_{21}B_{12} \oplus A_{22}B_{22} = I_n. & (6) \end{cases}$$

Applying both sides of each equation in the system to $x$ yields

$$\begin{cases} A_{12}B_{22}x = 0^n, & (7) \\ A_{22}B_{22}x = x. & (8) \end{cases}$$

Eq. (7) implies that $B_{22}x \in \ker(A_{12})$.

Now we demonstrate $x$ is the unique preimage in $\ker(B_{12})$ for image $B_{22}x \in \ker(A_{12})$. Assume there exists $x_1, x_2 \in \ker(B_{12})$ such that $B_{22}x_1 = B_{22}x_2$. Then

$$B_{22}(x_1 \oplus x_2) = 0^n. \tag{9}$$

From Eq. (8), we have

$$\left.\begin{array}{l} A_{22}B_{22}x_1 = x_1, \\ A_{22}B_{22}x_2 = x_2. \end{array}\right\} \Rightarrow A_{22}B_{22}(x_1 \oplus x_2) = x_1 \oplus x_2. \tag{10}$$

This yields $x_1 = x_2$ by

$$Eqs.\ (9) and\ (10) \Rightarrow (A_{22}0^n = x_1 \oplus x_2) \Rightarrow (0^n = x_1 \oplus x_2) \Rightarrow (x_1 = x_2).$$

Therefore, the mapping $x \mapsto B_{22}x$ is an injection from $\ker(B_{12})$ to $\ker(A_{12})$. Then $|\ker(B_{12})| \leq |\ker(A_{12})|$ and $\dim(\ker(B_{12})) \leq \dim(\ker(A_{12}))$.

(2) $\dim(\ker(A_{12})) \leq \dim(\ker(B_{12}))$: For $y \in \ker(A_{12})$, we can prove the mapping $y \mapsto A_{22}y$ is an injection from $\ker(A_{12})$ to $\ker(B_{12})$ from $\varphi^{-1}\varphi = I_{2n}$ as well.

Therefore, we obtain $\dim(\ker(A_{12})) = \dim(\ker(B_{12}))$. This yields $\mathrm{rank}(A_{12}) = \mathrm{rank}(B_{12})$. $\square$

## 3.2 Equivalence Between Parameters from Constructed Matrices

The transformations $\varphi$ and $\varphi^{-1}$ yield, respectively, the derived matrices $V_{A_{12},A_{22},i}$ and $V_{B_{12},B_{22},i}$ as given in Eq. (2). From these, the parameters $T(A_{12}, A_{22})$ and $T(B_{12}, B_{22})$ are computed according to the definition in Eq. (3). Likewise, the matrices $V_{A_{12}^\top,A_{11}^\top,i}$ and $V_{B_{12}^\top,B_{11}^\top,i}$ are obtained, leading to the parameters $T(A_{12}^\top, A_{11}^\top)$ and $T(B_{12}^\top, B_{11}^\top)$. These four parameters are crucial for determining the number of rounds required to achieve a pseudo-randomly secure UFLM construction. As will be shown in later sections, secure rounds depends primarily on $T(A_{12}, A_{22})$ and $T(A_{12}^\top, A_{11}^\top)$ for encryption, while secure rounds depends primarily on $T(B_{12}, B_{22})$ and $T(B_{12}^\top, B_{11}^\top)$ for decryption. In this section, we explore the relationships among these parameters. Our results reveal perfect symmetries between them. The main lemmas are stated as follows.

*Lemma 2* For any $\varphi, \varphi^{-1}$ defined in Eqs. (1) and (4), it holds that

$$T(A_{12}, A_{22}) = T(B_{12}, B_{22}).$$

This theorem is based on Lemma 1 and Eqs. (5) and (6) drived from $\varphi\varphi^{-1} = I_{2n}$. The detailed proof is presented in Section A.

*Lemma 3* For any $\varphi, \varphi^{-1}$ defined in Eqs. (1) and (4), it holds that

$$T(A_{12}^\top, A_{11}^\top) = T(B_{12}^\top, B_{11}^\top).$$

The equation $(\varphi^{-1}\varphi)^\top = (I_{2n})^\top$ leads to

$$\begin{cases} A_{22}^\top B_{12}^\top \oplus A_{12}^\top B_{11}^\top = O_n, \\ A_{21}^\top B_{12}^\top \oplus A_{11}^\top B_{11}^\top = I_n. \end{cases}$$

Based on Lemma 1 and the above equations, we conclude Lemma 3. The proof is omitted as it parallels the argument given for Lemma 2.

## 3.3 Impact of rank($A_{12}$) on the Parameter $T$

From the definition $T(A_{12}, A_{22}) = T(A_{12}^\top, A_{11}^\top)$ in Eq. (3), the following corollary is obtained directly.

*Corollary 1* For any $\varphi, \varphi^{-1}$ defined in Eqs. (1) and (4), if $\mathrm{rank}(A_{12}) = n$ then

$$T(A_{12}, A_{22}) = T(A_{12}^\top, A_{11}^\top) = 0.$$

*Corollary 2* For any $\varphi, \varphi^{-1}$ defined in Eqs. (1) and (4), if $\mathrm{rank}(A_{12}) < n$ then

$$T(A_{12}, A_{22}) \geq 1, T(A_{12}^\top, A_{11}^\top) \geq 1.$$

## 3.4 The Special Case: Involutory Linear Layers

For any involutory linear layer $\varphi$, both $V_{A_{12},A_{22},i}$ and $V_{A_{12}^\top,A_{11}^\top,i}$ exhibit rank stagnation at $\mathrm{rank}(A_{12})$ over all $i$, as stated in the lemma below.

*Lemma 4* For any linear layer $\varphi$ as defined in Eq. (1), if $\varphi^{-1} = \varphi$, then

$$\mathrm{rank}(V_{A_{12},A_{22},i}) = \mathrm{rank}(V_{A_{12}^\top,A_{11}^\top,i}) = \mathrm{rank}(A_{12})$$

holds for every non-negative integer $i$.

The proof is given in Section B. This lemma directly leads to the corollaries below.

*Corollary 3* For any linear layer $\varphi$ as defined in Eq. (1), if $\varphi^{-1} = \varphi$ and $\mathrm{rank}(A_{12}) = n$ then

$$T(A_{12}, A_{22}) = T(A_{12}^\top, A_{11}^\top) = 0.$$

*Corollary 4* For any linear layer $\varphi$ as defined in Eq. (1), if $\varphi^{-1} = \varphi$ and $\mathrm{rank}(A_{12}) < n$ then

$$T(A_{12}, A_{22}) = T(A_{12}^\top, A_{11}^\top) = \infty.$$

# 4 Differential Properties in $\mathbf{E}'_{[r]}$ and $\mathbf{D}'_{[r]}$

In this section, we present differential properties in $\mathbf{E}'_{[r]}$ and $\mathbf{D}'_{[r]}$. We first introduce two types of deterministic differential propagations for $\mathbf{E}'_{[r]}$ and $\mathbf{D}'_{[r]}$ respectively. The first type has zero left half of the output difference and propagates for at most $T(A_{12}, A_{22})$ rounds. The second type allows an arbitrary $n$-bit value as the left half of the output difference and propagates for one more round. These two properties will be used in CPA and CCA distinguishing attacks against UFLM as shown in Sections 6.1, 6.3, 7.1 and 7.2. We then prove that when the number of rounds is at least $T(A_{12}, A_{22}) + 1$, the probability of a collision between the outputs of $\mathbf{E}'_{[r]|L}$ or $\mathbf{D}'_{[r]|L}$ on two distinct inputs is bounded by $\frac{r}{2^{\mathrm{rank}(A_{12})}}$. Then prove that when the number of rounds is at least $T(A_{12}, A_{22}) + 2$, the probability of any difference between the outputs of $\mathbf{E}'_{[r]|L}$ or $\mathbf{D}'_{[r]|L}$ on two distinct inputs is bounded by $\frac{r}{2^{\mathrm{rank}(A_{12})}}$. These properties will be used later in the PRP and SPRP security proofs for UFLM in Sections 6.4 and 7.3.

## 4.1 Deterministic Differential Propagation for Collisions with up to $T(A_{12}, A_{22})$ Rounds

This section presents deterministic propagations with zero left-half output for the variants $\mathbf{E}'_{[r]}$ and $\mathbf{D}'_{[r]}$ when the number of rounds satisfies $r \leq T(A_{12}, A_{22})$, based on the ranges of $T(A_{12}, A_{22})$.

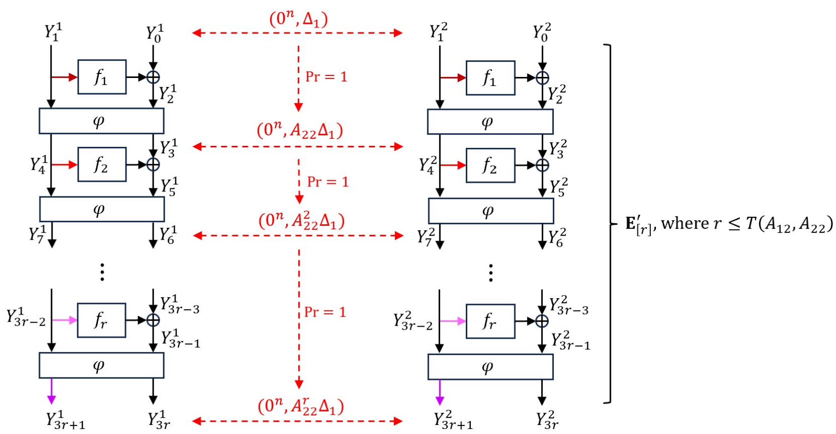



**Fig. 3**: Deterministic differential propagations in $\mathbf{E}'_{[r]}$ for $r \leq T(A_{12}, A_{22})$. Branches within the same color group share an identical value, whereas black branches do not necessarily share this property.

**Case 1: $\boldsymbol{T(A_{12}, A_{22}) \geq 1}$.** We first present the deterministic propagation with zero left-half output for $\mathbf{E}'_{[r]}$, as shown in Fig. 3. From the difitinon of $T(A_{12}, A_{22})$ as described in Eq. (3), we obtain that $r = T(A_{12}, A_{22}) + 1$ is the smallest number of rounds such that $V_{A_{12},A_{22},r-1}$ achieves full rank. Then for any number of rounds $r \leq T(A_{12}, A_{22})$, there always exists a non-zero $\Delta_1$ such that $V_{A_{12},A_{22},r-1}\Delta_1 = 0^n$, i.e.,

$$\begin{pmatrix} A_{12} \\ A_{12}A_{22} \\ A_{12}A_{22}^2 \\ \vdots \\ A_{12}A_{22}^{r-1} \end{pmatrix} \Delta_1 = \mathbf{0}.$$

Then for any two inputs $P^1 = (Y_1^1, Y_0^1)$ and $P^2 = (Y_1^2, Y_0^2)$ which satisfies that $P^1 \oplus P^2 = (0^n, \Delta_1)$, it follows that $F_1(Y_1^1) = F_1(Y_1^2), Y_2^1 \oplus Y_2^2 = Y_0^1 \oplus Y_0^2 = \Delta_1$ and for $i = 1$ to $r$:

$$\begin{aligned}
&Y_{3i-1}^1 \oplus Y_{3i-1}^2 = Y_{3i-3}^1 \oplus Y_{3i-3}^2 = A_{22}^{i-1}\Delta_1, \\
&Y_{3i}^1 \oplus Y_{3i}^2 = A_{22}(Y_{3i-1}^1 \oplus Y_{3i-1}^2) = A_{22}^i\Delta_1, \\
&Y_{3i+1}^1 \oplus Y_{3i+1}^2 = A_{12}(Y_{3i-1}^1 \oplus Y_{3i-1}^2) = A_{12}A_{22}^{i-1}\Delta_1 = 0^n \quad \Rightarrow F_i(Y_{3i+1}^1) = F_i(Y_{3i+1}^2)
\end{aligned}$$

This implies there is a deterministic differential propagation

$$(0^n, \Delta_1) \xrightarrow{\mathbf{E}'_{[r]}} (0^n, A_{22}^r\Delta_1). \tag{11}$$

Hence, for any plaintext pair with input difference $(0^n, \Delta_1)$, their left-half outputs after applying $\mathbf{E}'_{[r]}$ necessarily collide.

Given the structural symmetry between $\mathbf{E}'_{[r]}$ and $\mathbf{D}'_{[r]}$ and the equality $T(A_{12}, A_{22}) = T(B_{12}, B_{22})$ established in Lemma 2, the corresponding results for $\mathbf{D}'_{[r]}$ follow analogously. That is to say, for any number of rounds $r \leq T(A_{12}, A_{22})$, there always exists a non-zero $\Delta_2$ such that

$$(0^n, \Delta_2) \xrightarrow{\mathbf{D}'_{[r]}} (0^n, B_{22}^r\Delta_2). \tag{12}$$

Consequently, for every ciphertext pair with input difference $(0^n, \Delta_2)$, the left-half outputs after applying $\mathbf{D}'_{[r]}$ are identical.

**Case 2: $\boldsymbol{T(A_{12}, A_{22}) = 0}$.** In this case, only $r = 0$ satisfies $r \leq T(A_{12}, A_{22})$. We treat both the encryption and decryption processes of 0-round UFLM as identity transformations, i.e., $\mathbf{E}'_{[0]} = \mathbf{D}'_{[0]} \coloneqq I_{2n}$. Clearly, the deterministic differential propagations established for $T(A_{12}, A_{22}) \geq 1$ remain valid in this setting.

## 4.2 Deterministic Differential Propagation with up to $T(A_{12}, A_{22}) + 1$ Rounds

This section describes deterministic differential propagations for $\mathbf{E}'_{[r]}$ and $\mathbf{D}'_{[r]}$ with $r \leq T(A_{12}, A_{22}) + 1$. In contrast to Section 4.1, the left-half output difference here is not restricted to zero. Given that $\mathbf{E}'_{[r]}$ and $\mathbf{D}'_{[r]}$ already have deterministic differential propagations for $r \leq T(A_{12}, A_{22})$, we therefore directly present the deterministic differential propagation on $\mathbf{E}'_{[r]}$ and $\mathbf{D}'_{[r]}$ for the case $r = T(A_{12}, A_{22}) + 1$.

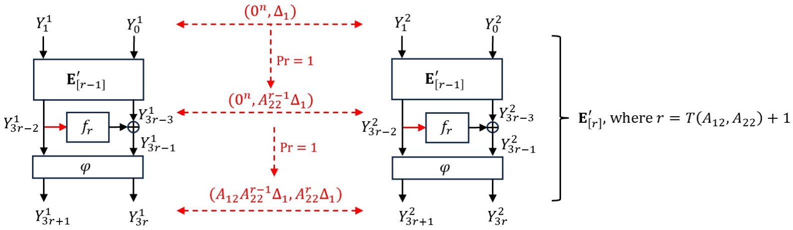



**Fig. 4**: Deterministic differential propagations in $\mathbf{E}'_{[r]}$ for $r = T(A_{12}, A_{22})+1$. Branches within the red color group share an identical value.

We first illustrate the deterministic differential propagation using $\mathbf{E}'_{[r]}$ as an example, as shown in Fig. 4. Let $\Delta_1$ be non-zero values such that $V_{A_{12},A_{22},r-2}\Delta_1 = 0^n$. According to Section 4.1, we obtain that

$$(0^n, \Delta_1) \xrightarrow{\mathbf{E}'_{[r-1]}} (0^n, A_{22}^{r-1}\Delta_1)$$

Obviously, it holds that

$$(0^n, A_{22}^{r-1}\Delta_1) \xrightarrow{\mathbf{R}_r} (A_{12}A_{22}^{r-1}\Delta_1, A_{22}^{r}\Delta_1)$$

where $A_{12}A_{22}^{r-1}\Delta_1 \neq 0^n$. Therefore, it follows directly that $\mathbf{E}'_{[r]} = \mathbf{R}_r \circ \mathbf{E}'_{[r-1]}$ admit the deterministic differential propagation

$$(0^n, \Delta_1) \xrightarrow{\mathbf{E}'_{[r]}} (A_{12}A_{22}^{r-1}\Delta_1,\ A_{22}^{r}\Delta_1). \tag{13}$$

This implies that for any plaintext pair with input difference $(0^n, \Delta_1)$, the left-half output difference after applying $\mathbf{E}'_{[r]}$ is invariably $A_{12}A_{22}^{r-1}\Delta_1$.

The structural symmetry between $\mathbf{E}'_{[r]}$ and $\mathbf{D}'_{[r]}$, together with Lemma 2, yields analogous results for $\mathbf{D}'_{[r]}$. Specifically, a non-zero $\Delta_2$ exists such that

$$(0^n, \Delta_2) \xrightarrow{\mathbf{D}'_{[r]}} (B_{12}B_{22}^{r-1}\Delta_2, B_{22}^{r}\Delta_2), \tag{14}$$

where $B_{12}B_{22}^{r-1}\Delta_2 \neq 0^n$. Hence, for every ciphertext pair with input difference $(0^n, \Delta_2)$, the left-half output difference after applying $\mathbf{D}'_{[r]}$ is always $B_{12}B_{22}^{r-1}\Delta_2$.

### 4.3 Bounding Probability for Collision for $T(A_{12}, A_{22}) + 1$ Rounds and Beyond

In this section, we prove that when the number of rounds is at least $T(A_{12}, A_{22}) + 1$, the probability of a collision between the outputs of $\mathbf{E}'_{[r]|L}$ (resp. $\mathbf{D}'_{[r]|L}$) on two distinct inputs is bounded by $\frac{r}{2^{\mathrm{rank}(A_{12})}}$. A formal description follows.

*Lemma 5* Let $\mathbf{E}'_{[r]}$ be encryption variant of $r$-round UFLM over $2n$-bit input, where $r \geq T(A_{12}, A_{22}) + 1$ and $T(A_{12}, A_{22}) < \infty$. Then, for any two distinct inputs $P$ and $P'$, the probability that their left outputs collide is bounded by

$$\Pr_{f_1,\ldots,f_r}\left[\mathbf{E}'_{[r]|L}(P) = \mathbf{E}'_{[r]|L}(P')\right] \leq \frac{r}{2^{\mathrm{rank}(A_{12})}}.$$

*Proof* Represent $P = (Y_1, Y_0), P' = (Y'_1, Y'_0) \in \{0,1\}^n \times \{0,1\}^n$, which are any two distinct inputs. Given $r \geq T(A_{12}, A_{22}) + 1$ with $T(A_{12}, A_{22}) < \infty$, there always exists a smallest index $j \in \{1, 2, \ldots, T(A_{12}, A_{22}) + 1\}$ such that $Y_{3j-2} \neq Y'_{3j-2}$.

We firstly consider the case that $j \in \{1, 2, \ldots, T(A_{12}, A_{22})\}$. For $Y_{3j-2} \neq Y'_{3j-2}$, we compute the probability of the event $Y_{3j+1} = Y'_{3j+1}$. Note that $Y_{3j+1} = f_j(Y_{3j-2}) \oplus Y_{3j-1}$, then

$$\begin{aligned}
&\Pr_{f_j \leftarrow\$\mathrm{Func}(n,n)}[Y_{3j+1} = Y'_{3j+1}] \\
=&\Pr_{f_j \leftarrow\$\mathrm{Func}(n,n)}[Y_{3j+1} = Y'_{3j+1} \mid Y_{3j-2} \neq Y'_{3j-2}] \\
=&\Pr_{f_j \leftarrow\$\mathrm{Func}(n,n)}\left[\begin{array}{l} A_{11}Y_{3j-2} \oplus A_{12}(f_j(Y_{3j-2}) \oplus Y_{3j-1}) \\ = A_{11}Y'_{3j-2} \oplus A_{12}(f_j(Y'_{3j-2}) \oplus Y'_{3j-1}) \end{array} \middle| Y_{3j-2} \neq Y'_{3j-2}\right] \\
=&\Pr_{f_j \leftarrow\$\mathrm{Func}(n,n)}\left[\begin{array}{l} f_j(Y_{3j-2}) \oplus f_j(Y'_{3j-2}) \\ \in \left\{ \begin{array}{l} A_{11}(Y_{3j-2} \oplus Y'_{3j-2}) \oplus \\ A_{12}(Y_{3j-1} \oplus Y'_{3j-1}) \oplus Z \end{array} \middle| Z \in \ker(A_{12}) \right\} \end{array} \middle| Y_{3j-2} \neq Y'_{3j-2}\right] \\
\leq&\frac{2^{n(2^n-2)} \cdot 2^n \cdot 2^{n-\mathrm{rank}(A_{12})}}{2^{n2^n}} = \frac{1}{2^{\mathrm{rank}(A_{12})}}.
\end{aligned}$$

For the subsequent pairs $(Y_{3j+1}, Y_{3j})$ and $(Y'_{3j+1}, Y'_{3j})$, we compute the probability of the event $Y_{3j+4} = Y'_{3j+4}$:

$$\begin{aligned}
&\Pr_{f_j, f_{j+1} \leftarrow\$\mathrm{Func}(n,n)}[Y_{3j+4} = Y'_{3j+4}] \\
=&\Pr_{f_{j+1} \leftarrow\$\mathrm{Func}(n,n)}[Y_{3j+4} = Y'_{3j+4} \mid Y_{3j+1} \neq Y'_{3j+1}] \cdot \Pr_{f_j \leftarrow\$\mathrm{Func}(n,n)}[Y_{3j+1} \neq Y'_{3j+1}] \\
&+ \Pr_{f_{j+1} \leftarrow\$\mathrm{Func}(n,n)}[Y_{3j+4} = Y'_{3j+4} \mid Y_{3j+1} = Y'_{3j+1}] \cdot \Pr_{f_j \leftarrow\$\mathrm{Func}(n,n)}[Y_{3j+1} = Y'_{3j+1}] \\
\leq&\frac{1}{2^{\mathrm{rank}(A_{12})}}\left(1 - \Pr_{f_j \leftarrow\$\mathrm{Func}(n,n)}[Y_{3j+1} = Y'_{3j+1}]\right) + 1 \cdot \Pr_{f_j \leftarrow\$\mathrm{Func}(n,n)}[Y_{3j+1} = Y'_{3j+1}]
\end{aligned}$$

$$\leq \frac{1}{2^{\mathrm{rank}(A_{12})}}\left(1-\frac{1}{2^{\mathrm{rank}(A_{12})}}\right)+1\cdot\frac{1}{2^{\mathrm{rank}(A_{12})}}$$
$$\leq \frac{2}{2^{\mathrm{rank}(A_{12})}}.$$

By mathematical induction, we conclude that the probability of a collision for the left-half outputs through $\mathbf{E}'_{[r]|_L}$ is:

$$\Pr_{f_j,\ldots,f_r \leftarrow\$\mathrm{Func}(n,n)}[Y_{3r+1}=Y'_{3r+1}] \leq \frac{r-j+1}{2^{\mathrm{rank}(A_{12})}}.$$

Since $j \in \{1,2,\ldots,T(A_{12},A_{22})\}$, the above probability is bound by $\frac{r}{2^{\mathrm{rank}(A_{12})}}$.

For $j=T(A_{12},A_{22})+1$, if $r=T(A_{12},A_{22})+1=j$, we obtain $Y_{3r+1}=Y'_{3r+1}$ with probbaility 0 directly; else, $\Pr_{f_j,\ldots,f_r \leftarrow\$\mathrm{Func}(n,n)}[Y_{3r+1}=Y'_{3r+1}] \leq \frac{r-T(A_{12},A_{22})}{2^{\mathrm{rank}(A_{12})}} \leq \frac{r}{2^{\mathrm{rank}(A_{12})}}$.

Consequently, $\Pr_{f_1,\ldots,f_r}\left[\mathbf{E}'_{[r]|_L}(P)=\mathbf{E}'_{[r]|_L}(P')\right] \leq \frac{r}{2^{\mathrm{rank}(A_{12})}}$ holds. □

Building on Lemma 5 for $\mathbf{E}'_{[r]}$, we establish a corresponding claim for $\mathbf{D}'_{[r]}$. Together with Lemmas 1 and 2, this leads to the following result.

*Lemma 6* Let $\mathbf{D}'_{[r]}$ be encryption variant of $r$-round UFLM over $2n$-bit input, where $r \geq T(A_{12},A_{22})+1$ and $T(A_{12},A_{22})<\infty$. Then, for any two distinct inputs $C$ and $C'$, the probability that their left outputs collide is bounded by

$$\Pr_{f_1,\ldots,f_r}\left[\mathbf{D}'_{[r]|_L}(C)=\mathbf{D}'_{[r]|_L}(C')\right] \leq \frac{r}{2^{\mathrm{rank}(A_{12})}}.$$

## 4.4 Bounding Probability for Any Output Difference for $T(A_{12}, A_{22})+2$ Rounds and Beyond

In this section, we prove that when the number of rounds is at least $T(A_{12},A_{22})+2$, the probability of any difference between the outputs of $\mathbf{E}'_{[r]|L}$ (resp. $\mathbf{D}'_{[r]|L}$) on two distinct inputs is bounded by $\frac{r}{2^{\mathrm{rank}(A_{12})}}$. A formal description follows.

*Lemma 7* Let $\mathbf{E}'_{[r]}$ be encryption variant of $r$-round UFLM over $2n$-bit input, where $r \geq T(A_{12},A_{22})+2$ and $T(A_{12},A_{22})<\infty$. Then, for any two distinct inputs $P$ and $P'$ and any left-half output difference $\Delta \in \{0,1\}^n$,

$$\Pr_{f_1,\ldots,f_r}\left[\mathbf{E}'_{[r]|_L}(P)\oplus\mathbf{E}'_{[r]|_L}(P')=\Delta\right] \leq \frac{r}{2^{\mathrm{rank}(A_{12})}}.$$

*Proof* Represent $P=(Y_1,Y_0), P'=(Y'_1,Y'_0) \in \{0,1\}^n \times \{0,1\}^n$, which are any two distinct inputs. For any $r \geq T(A_{12},A_{22})+2$ and $T(A_{12},A_{22})<\infty$, from Lemma 5 we obtain that $\Pr_{f_1,\ldots,f_{r-1}}\left[Y_{3r-2}=Y'_{3r-2}\right] \leq \frac{r-1}{2^{\mathrm{rank}(A_{12})}}$. We compute the probability of the event $Y_{3r+1}\oplus Y'_{3r+1}=\Delta$:

$$\Pr_{f_1,\ldots,f_r \leftarrow\$\mathrm{Func}(n,n)}[Y_{3r+1}\oplus Y'_{3r+1}=\Delta]$$
$$=\Pr_{f_r \leftarrow\$\mathrm{Func}(n,n)}[Y_{3r+1}\oplus Y'_{3r+1}=\Delta \mid Y_{3r-2}\neq Y'_{3r-2}]\cdot \Pr_{f_1,\ldots,f_{r-1} \leftarrow\$\mathrm{Func}(n,n)}[Y_{3r-2}\neq Y'_{3r-2}]$$

$$
\begin{aligned}
&+ \Pr_{f_r \leftarrow \$\mathrm{Func}(n,n)}[Y_{3r+1} \oplus Y'_{3r+1} = \Delta \mid Y_{3r-2} = Y'_{3r-2}] \cdot \Pr_{f_1,\ldots,f_{r-1} \leftarrow \$\mathrm{Func}(n,n)}[Y_{3r-2} = Y'_{3r-2}] \\
\leq & \frac{1}{2^{\mathrm{rank}(A_{12})}} \left(1 - \Pr_{f_1,\ldots,f_{r-1} \leftarrow \$\mathrm{Func}(n,n)}[Y_{3r-2} = Y'_{3r-2}]\right) \\
&+ 1 \cdot \Pr_{f_1,\ldots,f_{r-1} \leftarrow \$\mathrm{Func}(n,n)}[Y_{3r-2} = Y'_{3r-2}] \\
\leq & \frac{1}{2^{\mathrm{rank}(A_{12})}} \left(1 - \frac{r-1}{2^{\mathrm{rank}(A_{12})}}\right) + 1 \cdot \frac{r-1}{2^{\mathrm{rank}(A_{12})}} \\
\leq & \frac{r}{2^{\mathrm{rank}(A_{12})}},
\end{aligned}
$$

where $Y_{3r+1}$ and $Y'_{3r+1}$ are left-half outputs through $\mathbf{E}'_{[r]|_L}$. Thus, the proof is concluded. □

By extending Lemma 7 from $\mathbf{E}'_{[r]}$ to $\mathbf{D}'_{[r]}$ and invoking Lemmas 1 and 2, we obtain the following result.

*Lemma 8* Let $\mathbf{D}'_{[r]}$ be encryption variant of $r$-round UFLM over $2n$-bit input, where $r \geq T(A_{12}, A_{22}) + 2$ and $T(A_{12}, A_{22}) < \infty$. Then, for any two distinct inputs $C$ and $C'$ and any left-half output difference $\Delta \in \{0,1\}^n$,

$$
\Pr_{f_1,\ldots,f_r}\left[\mathbf{D}'_{[r]|_L}(C) \oplus \mathbf{D}'_{[r]|_L}(C') = \Delta\right] \leq \frac{r}{2^{\mathrm{rank}(A_{12})}}.
$$

# 5 PRP Security of $\mathbf{E}_{[r]}$ and $\mathbf{D}_{[r]}$ under Constraints

In this section, we show the PRP security of $\mathbf{E}_{[r]}$ and $\mathbf{D}_{[r]}$ under constraints. First, for $\mathbf{E}_{[r]}$ and similarly for $\mathbf{D}_{[r]}$, if the inputs to each round function are distinct across different queries, then we show that $\mathbf{E}_{[r]}$ and $\mathbf{D}_{[r]}$ behave as pseudorandom functions once the number of rounds reaches $T(A_{12}^\top, A_{11}^\top) + 2$. This property will be used later in the PRP and SPRP security proofs for UFLM in Sections 6.4 and 7.3. Second, when the left halves of the inputs may coincide, we show that $\mathbf{E}_{[r]}$ and $\mathbf{D}_{[r]}$ with output projection exhibit deterministic differential propagations provided the number of rounds at most $T(A_{12}^\top, A_{11}^\top)+2$. This property will be used in the CPA distinguishing attack against UFLM as shown in Section 6.3.

## 5.1 Achieving PRP Security at $T(A_{12}^\top, A_{11}^\top) + 2$ Rounds with Distinct Round-Function Inputs

We first consider the PRP security for $\mathbf{E}_{[r]}$ with distinct round-function inputs when $r = T(A_{12}^\top, A_{11}^\top) + 2$. The right half of the output of $\mathbf{E}_{[r]}$, $Y_{3r-1} = f_r(Y_{3r-2}) \oplus Y_{3r-3}$, is inherently nonlinear due to the round function $f_r$. And the left output half $Y_{3r-2}$ also can be expressed in terms of the nonlinear outputs $Y_{3i-1} = f_i(Y_{3i-2}) \oplus Y_{3i-3}$ for $i = 1, \ldots, r-1$ and the initial left half $Y_1$ as follows:

$$
Y_{3r-2} = A_{11}^{r-1} Y_1 \oplus \bigoplus_{i=1}^{r-1} A_{11}^{r-1-i} A_{12} Y_{3i-1}
$$

$$
\begin{aligned}
&= A_{11}^{r-1} Y_1 \oplus \begin{pmatrix} A_{12} & A_{11}A_{12} & \cdots & A_{11}^{r-2}A_{12} \end{pmatrix} \begin{pmatrix} Y_{3r-4} \\ Y_{3r-7} \\ \vdots \\ Y_2 \end{pmatrix} \\
&= A_{11}^{r-1} Y_1 \oplus \begin{pmatrix} A_{12}^\top \\ A_{12}^\top A_{11}^\top \\ \vdots \\ A_{12}^\top \left(A_{11}^\top\right)^{r-2} \end{pmatrix}^\top \begin{pmatrix} Y_{3r-4} \\ Y_{3r-7} \\ \vdots \\ Y_2 \end{pmatrix} = A_{11}^{r-1} Y_1 \oplus \left(V_{A_{12}^\top, A_{11}^\top, r-2}\right)^\top Y
\end{aligned} \tag{15}
$$

where $Y^\top = \begin{pmatrix} Y_{3r-4}^\top & Y_{3r-7}^\top & \ldots & Y_5^\top & Y_2^\top \end{pmatrix}$, and $V_{A_{12}^\top, A_{11}^\top, r-2}$ is defined as in Eq. (2).

According to the definition of $T(A_{12}^\top, A_{11}^\top)$ in Eq. (3), $r = T(A_{12}^\top, A_{11}^\top) + 2$ is the smallest number of rounds such that matrix $V_{A_{12}^\top, A_{11}^\top, r-2}$ achieves full rank. For each round function $f_i$, if its inputs remain distinct across all calls, then the nonlinear output values $Y_2, Y_5, \ldots, Y_{3r-4}, Y_{3r-1}$ can be regarded as random, i.e., the vector $Y$ is random. A full-column-rank matrix then acts on this random vector $Y$ to produce the left-half output $Y_{3r-2}$; hence $Y_{3r-2}$ inherits randomness. The right-half output directly equals $Y_{3r-1}$ and therefore also possesses randomness. In this way, the entire output can be shown to be random. The formal lemma and its proof are given below.

*Lemma 9* Let $\mathbf{E}_{[r]}$ be the encryption algorithm of $r$-round UFLM with $r = T(A_{12}^\top, A_{11}^\top) + 2$, where $T(A_{12}^\top, A_{11}^\top) < \infty$. Let $\mathcal{A}$ be any PRP adversary on $\mathbf{E}_{[r]}$ that submits at most $q$ encryption queries. For each round function $f_i$, if its inputs remain distinct across all calls, then

$$
\mathrm{Adv}_{\mathbf{E}_{[r]}}^{\mathrm{prp}}(\mathcal{A}) \leq \frac{0.5q^2}{2^{2n}}.
$$

*Proof* Let the intermediate values are recorded as $Y_j^i$, which denote the contents of the $Y_j$ registers, respectively, during the $i$-th query. The ciphertext $C^i$ can be represented as

$$
C^i = \begin{pmatrix} Y_{3r-2}^i \\ Y_{3r-1}^i \end{pmatrix} = \begin{pmatrix} A_{11} Y_1^i \\ 0^n \end{pmatrix} \oplus \widetilde{V}_r \widetilde{Y}^i
$$

where

$$
\widetilde{V} = \begin{pmatrix} I_n & O_{n\times(r-1)n} \\ O_n & \left(V_{A_{12}^\top, A_{11}^\top, r-2}\right)^\top \end{pmatrix}, (\widetilde{Y}^i)^\top = \begin{pmatrix} (Y_{3r-1}^i)^\top & (Y_{3r-4}^i)^\top & \ldots & (Y_{3j-1}^i) & \ldots & (Y_2^i)^\top \end{pmatrix}.
$$

From $r = T(A_{12}^\top, A_{11}^\top) + 2$, we obtain $\widetilde{V}$ is full rank. Consider an arbitrary ciphertext $C^i \in \{0,1\}^{2n}$. The requirement that the internal variables $\widetilde{Y}^i$ must map $P^i$ to $C^i$ imposes $2n$ linear constraints on the $rn$-bit vector $\widetilde{Y}^i$. Since $\widetilde{V}$ is full rank, the set of valid intermediate variable vectors $\widetilde{Y}^i$ forms an affine subspace of dimension $rn - 2n$. In other words, there are exactly $2^{rn-2n}$ distinct sequences of internal values compatible with $(P^i, C^i)$. Since for each round function $f_i$, if its inputs remain distinct across all calls, for any single valid sequence of internal values, the probability that these specific outputs $Y_2^i, Y_5^i, \ldots, Y_{3r-1}^i$ are produced by the random functions $f_1, \ldots, f_r$ is $\frac{1}{2^{rn}}$. Therefore, summing over all valid sequences, the probability for a query to get any $C^i \in \{0,1\}^{2n}$ is $\frac{1}{2^{2n}}$ by the uniform random functions $f_1, f_2, \ldots, f_r$. This is equivalent to sampling each $C^i$ from $\{0,1\}^{2n}$ uniformly at random. The only deviation from a uniformly random permutation lies in the possibility of output collisions among the queries. Based on PRP-PRF switching lemma [19, 20], after $q$ encryption queries, the distinguishing advantage between the two is bounded above by $\frac{0.5q^2}{2^{2n}}$. □

Building on Lemma 9 for $\mathbf{E}_{[r]}$, an analogous result holds for $\mathbf{D}_{[r]}$. Combining this with Lemma 3 and PRP-PRF switching lemma yields the following lemma.

*Lemma 10* Let $\mathbf{D}_{[r]}$ be the decryption algorithm of $r$-round UFLM with $r = T(A_{12}^\top, A_{11}^\top)+2$, where $T(A_{12}^\top, A_{11}^\top) < \infty$ and distinct left-half inputs. Let $\mathcal{A}$ be any PRP adversary on $\mathbf{D}_{[r]}$ that submits at most $q$ encryption queries. For each round function $f_i$, if its inputs remain distinct across all calls, then

$$\mathrm{Adv}_{\mathbf{D}_{[r]}}^{\mathrm{prp}}(\mathcal{A}) \leq \frac{0.5q^2}{2^{2n}}.$$

## 5.2 Deterministic Differential Propagation through UFLM with Output Projection under Zero Left-Half Input Difference up to $T\left(A_{12}^\top, A_{11}^\top\right)+2$ Rounds

For a round count of $r \leq T(A_{12}^\top, A_{11}^\top)+2$, there are deterministic differential propagations through $\mathbf{E}_{[r]}$ and $\mathbf{D}_{[r]}$ with output projections under collisions of the left-half inputs. We illustrate the attack using $\mathbf{E}_{[r]}$ as an example by categorizing the ranges of $T(A_{12}^\top, A_{11}^\top)$, as shown in Fig. 5. The attack on $\mathbf{D}_{[r]}$ is entirely analogous due to the structural symmetry between the two and is therefore omitted.

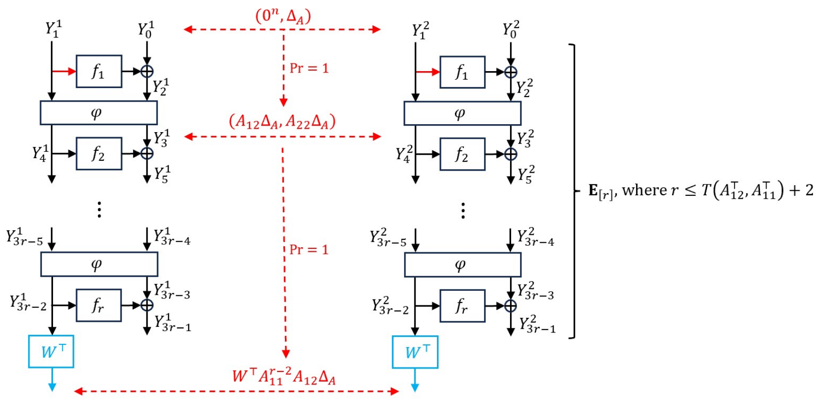


**Fig. 5**: Deterministic differential propagations in $W^\top \circ \mathbf{E}_{[r]}$ for $r \leq T(A_{12}^\top, A_{11}^\top)+2$. Branches within the red color group share an identical value.

**Case 1: $T(A_{12}^\top, A_{11}^\top) \geq 1$.** We decompose $\mathbf{E}_{[r]} = \mathbf{E}_{[2,r]} \circ \mathbf{R}_1$. It is straightforward to obtain the deterministic differential propagation

$$(0^n, \Delta_A) \xrightarrow{\mathbf{R}_1} (A_{12}\Delta_A, A_{22}\Delta_A)$$

for the $\mathbf{R}_1$ stage. Then we focus on $\mathbf{E}_{[2,r]}$. The left part of the ciphertext can be represented by

$$Y_{3r-2} = A_{11}^{r-2} Y_4 \oplus \bigoplus_{i=2}^{r-1} A_{11}^{r-1-i} A_{12} Y_{3i-1} = A_{11}^{r-2} Y_4 \oplus (V_{A_{12}^\top, A_{11}^\top, r-3})^\top Y \tag{16}$$

where $Y^\top = \begin{pmatrix} Y_{3r-4}^\top & Y_{3r-7}^\top & \dots & Y_5^\top \end{pmatrix}$. The matrix $V_{A_{12}^\top, A_{11}^\top, r-3}$ is rank-deficient because $r - 3 < T(A_{12}^\top, A_{11}^\top)$, whereas $j = T(A_{12}^\top, A_{11}^\top)$ is the smallest index for which $V_{A_{12}^\top, A_{11}^\top, j}$ achieves full rank. We abbreviate $\mathrm{rank}(V_{A_{12}^\top, A_{11}^\top, r-3})$ as $\mathrm{rank}(V_3)$. Then there exists an $n \times (n - \mathrm{rank}(V_3))$ matrix $W$ of rank $n - \mathrm{rank}(V_3)$ such that $V_{A_{12}^\top, A_{11}^\top, r-3} W = O_{(r-2)n \times (n-\mathrm{rank}(V_3))}$. Thus, applying $W^\top$ to $Y_{3r-2}$ in Eq. (16) gives $W^\top Y_{3r-2} = W^\top A_{11}^{r-2} Y_4$. Hence, $W^\top$ removes the effect of the variable $Y$ and the output depends only on the left half $Y_4$ of the input to $\mathbf{E}_{[2,r]}$. Then $W^\top \circ \mathbf{E}_{[2,r]|L}$ admits the deterministic differential propagation

$$(A_{12}\Delta_A,\ A_{22}\Delta_A) \xrightarrow{W^\top \circ \mathbf{E}_{[2,r]|L}} W^\top A_{11}^{r-2} A_{12} \Delta_A.$$

Therefore, $W^\top \circ \mathbf{E}_{[r]|L}$ admits the deterministic differential propagation

$$(0^n, \Delta_A) \xrightarrow{W^\top \circ \mathbf{E}_{[r]|L}} W^\top A_{11}^{r-2} A_{12} \Delta_A.$$

**Case 2: $T(A_{12}^\top, A_{11}^\top) = 0$.** In this case, $r \leq 2$. With $W \coloneqq I_n$, it follows directly that $W^\top \circ \mathbf{E}_{[r]|L}$ also admits the deterministic differential propagations

$$\begin{aligned}
(0^n, \Delta_A) &\xrightarrow{W^\top \circ \mathbf{E}_{[r]|L}} A_{12}\Delta_A, r = 2, \\
(0^n, \Delta_A) &\xrightarrow{W^\top \circ \mathbf{E}_{[r]|L}} 0^n, r = 1.
\end{aligned}$$

# 6 CPA Security Analysis of UFLM

In this section, we present the CPA security analysis of UFLM. When $T(A_{12}, A_{22}) = \infty$ or $T(A_{12}^\top, A_{11}^\top) = \infty$, a distinguishing attack exists with at most two encryption queries. Otherwise, if the number of rounds is less than $T(A_{12}, A_{22}) + T(A_{12}^\top, A_{11}^\top) + 2$, a distinguishing attack also exists with only two encryption queries. When the number of rounds exceeds $T(A_{12}, A_{22}) + T(A_{12}^\top, A_{11}^\top) + 3$, PRP security is achieved.

## 6.1 CPA Distinguishers for an Arbitrary Number of Rounds under $T(A_{12}, A_{22}) = \infty$

The condition $T(A_{12}, A_{22}) = \infty$ implies there exists a non-zero $\Delta_1$ such that $A_{12}A_{22}^i\Delta_1 = 0^n$ for any $i$. As analyzed in Section 4.1, for an input difference $(0^n, \Delta_1)$ with such $\Delta_1$, the left-half output difference of $\mathbf{E}'[r]$ remains $0^n$ for any number of rounds; the same holds for $\mathbf{E}[r]$. Consequently, it can be distinguished from a random permutation using only two encryption queries with probability almost 1.

## 6.2 CPA Distinguishers for an Arbitrary Number of Rounds under $T(A_{12}^\top, A_{11}^\top) = \infty$

When $T(A_{12}^\top, A_{11}^\top) = \infty$, the matrix $V_{A_{12}^\top, A_{11}^\top, r-2}$ in the expression for the left output $Y_{3r-2}$ (see Eq. (15)) is not full rank for any number of rounds. Hence there exists a matrix $W$ with rank $n - \mathrm{rank}(V_{A_{12}^\top, A_{11}^\top, r-2})$ such that $V_{A_{12}^\top, A_{11}^\top, r-2}W = O$. Applying $W^\top$ to the left output $Y_{3r-2}$ yields $W^\top A_{11}^{r-1}Y_1$; however, this relation holds for random permutation only with probability $\frac{1}{2^{n-\mathrm{rank}(V_2)}}$, where $\mathrm{rank}(V_2)$ abbreviates $\mathrm{rank}\big(V_{A_{12}^\top, A_{11}^\top, r-2}\big)$. Consequently, the CPA distinguishing attack succeeds with probability $1 - \frac{1}{2^{n-\mathrm{rank}(V_2)}} \geq \frac{1}{2}$ using merely one query.

## 6.3 CPA Distinguishers for up to $T(A_{12}, A_{22}) + T(A_{12}^\top, A_{11}^\top) + 2$ Rounds

This section presents a CPA distinguishing attack on UFLM when the number of rounds satisfies $r \leq T(A_{12}, A_{22}) + T(A_{12}^\top, A_{11}^\top) + 2$, where $T(A_{12}, A_{22}) < \infty, T(A_{12}^\top, A_{11}^\top) < \infty$.

Let

$$r_1 = \begin{cases} r - T(A_{12}^\top, A_{11}^\top) - 2, & \text{if } T(A_{12}^\top, A_{11}^\top) + 2 \leq r \leq T(A_{12}, A_{22}) + T(A_{12}^\top, A_{11}^\top) + 2, \\ 0, & \text{if } r < T(A_{12}^\top, A_{11}^\top) + 2, \end{cases}$$

and $r_2 = r - r_1$. Decompose $\mathbf{E}_{[r]} = \mathbf{E}_{[r_1+1,r]} \circ \mathbf{E}'_{[r_1]} = \mathbf{E}_{[r-r_2+1,r]} \circ \mathbf{E}'_{[r_1]}$. And then we will show that $W^\top \circ \mathbf{E}_{[r]|L}$ has a deterministic difference propagation, as illustrated in Fig. 6.

The first part $\mathbf{E}'_{[r_1]}$ uses at most $T(A_{12}, A_{22})$ rounds. From Section 4.1, $\mathbf{E}'_{[r_1]}$ possesses the deterministic differential propagation

$$(0^n, \Delta_1) \xrightarrow{\mathbf{E}'_{[r_1]}} (0^n, A_{22}^{r_1}\Delta_1),$$

for some nonzero $\Delta_1 \in \{0,1\}^n$.

The second part $\mathbf{E}_{[r-r_2+1,r]}$ uses at most $T(A_{12}^\top, A_{11}^\top) + 2$ rounds. From Section 5.2, for any input difference whose left half is zero there exists a matrix $W$ of rank at least 1 such that $W^\top \circ \mathbf{E}_{[r-r_2+1,r]|L}$ admits the deterministic differential propagation

$$(0^n, A_{22}^{r_1}\Delta_1) \xrightarrow{W^\top \circ \mathbf{E}_{[r-r_2+1,r]|L}} W^\top A_{11}^{r_2-2}A_{12}A_{22}^{r_1}\Delta_1.$$

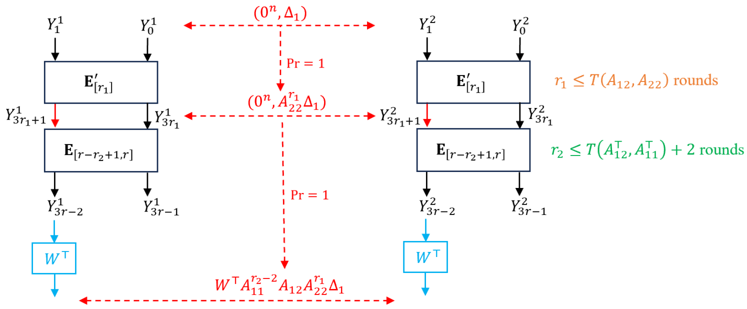



**Fig. 6**: Deterministic differential propagations in $W^\top \circ \mathbf{E}_{[r]}$ for $r \leq T(A_{12}, A_{22}) + T(A_{12}^\top, A_{11}^\top) + 2$.

Concatenating the two propagations yields the deterministic differential propagation for the composite map:

$$(0^n, \Delta_1) \xrightarrow{W^\top \circ \mathbf{E}_{[r]|L}} W^\top A_{11}^{r_2-2} A_{12} A_{22}^{r_1} \Delta_1.$$

Then we can make a successful CPA distinguishing attack as follows.

(1) Randomly choose a pair of plaintexts $(P^1, P^2)$ that satisfies $P^1 \oplus P^2 = (0^n, \Delta_1)$, and then query the encryption oracle to obtain a ciphertexts pair $(C^1, C^2)$.

(2) If $W^\top({C^1}_{|L} \oplus {C^2}_{|L}) = W^\top A_{11}^{r_2-2} A_{12} A_{22}^{r_1} \Delta_1$, output 1; else, output 0.

For the $\mathbf{E}_{[r]}$, this attack always yields a value of 1. For a random permutation, because $C^1$ and $C^2$ are outputs of distinct inputs and $W$ has rank at least 1, the probability that $W^\top\left({C^1}_{|L} \oplus {C^2}_{|L}\right) = W^\top A_{11}^{r_2-2} A_{12} A_{22}^{r_1} \Delta_1$ holds is at most $\frac{1}{2}$. Hence, the distinguishing advantage between $\mathbf{E}_{[r]}$ and a random permutation is at least $\frac{1}{2}$.

### 6.4 CPA Security for $T(A_{12}, A_{22}) + T(A_{12}^\top, A_{11}^\top) + 3$ Rounds

This section establishes the PRP security of UFLM for $r = T(A_{12}, A_{22}) + T(A_{12}^\top, A_{11}^\top) + 2$ rounds, assuming $T(A_{12}, A_{22}) < \infty$ and $T(A_{12}^\top, A_{11}^\top) < \infty$. The formal statement follows.

**Theorem 1** *Let $\mathbf{E}_{[r]}$ be the encryption algorithm of $r$-round UFLM with $r = T(A_{12}, A_{22}) + T(A_{12}^\top, A_{11}^\top) + 3$, where $T(A_{12}, A_{22}) < \infty, T(A_{12}^\top, A_{11}^\top) < \infty$. Let $\mathcal{A}$ be any PRP adversary against $\mathbf{E}_{[r]}$ that makes at most $q$ encryption queries. Then*

$$\mathrm{Adv}^{\mathrm{prp}}_{\mathbf{E}_{[r]}}(\mathcal{A}) \leq \frac{0.5q^2r^2}{2^{\mathrm{rank}(A_{12})}} + \frac{0.5q^2}{2^{2n}}.$$

*Proof* Let $r_1 := T(A_{12}, A_{22}) + 1$ and $r_2 := T(A_{12}^\top, A_{11}^\top) + 2$. Then $r = r_1 + r_2$ and $\mathbf{E}_{[r]} = \mathbf{E}_{[r-r_2+1,r]} \circ \mathbf{E}'_{[r_1]}$. The following proof proceeds in two steps. First, for any $c \geq r_1$, the collision probability of $\mathbf{E}'_{[c]|L}$ outputs is bounded when $\mathbf{E}_{[r]}$ inputs differ; equivalently, the inputs to each round function $f_i$ in $\mathbf{E}_{[r-r_2+1,r]}$ collide across queries with bounded probability. Second, conditioned on no such collisions, $\mathbf{E}_{[r-r_2+1,r]}$ is a PRP, hence $\mathbf{E}_{[r]}$ is a PRP. This argument is illustrated in Fig. 7. For $q$ encryption queries to UFLM, we denote by $P^i$ the plaintext and $Y_j^i$ the value of the variable $Y_j$ in the $i$-th query. We require no repeated queries.

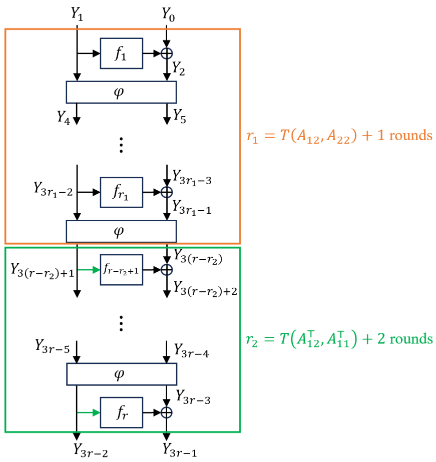



**Fig. 7**: Illustration of the CPA proof for UFLM when the number of rounds satisfies $r = T(A_{12}, A_{22}) + T(A_{12}^\top, A_{11}^\top) + 3$. Branches in green correspond to bounded collision probabilities.

Firstly, to capture the collisions among the inputs to any of the round functions $f_{r-r_2+1}, \ldots, f_r$ across queries in $\mathbf{E}_{[r-r_2+1,r]}$, we define the bad event as follows.

- Bad: $Y^j_{3(r-r_2+k)-2} = Y^i_{3(r-r_2+k)-2}$ for some $1 \le j < i \le q, k \in \{1, 2, \ldots, r_2\}$.

$Y_{3(r-r_2+k)-2}$ is the outputs of $\mathbf{E}'_{[r-r_2+k-1]|L}$, where the number of round $r - r_2 + k - 1 \ge r_1 = T(A_{12}, A_{22}) + 1$. Then by Lemma 5,

$$\begin{aligned}&\Pr\big[Y^j_{3(r-r_2+k)-2} = Y^i_{3(r-r_2+k)-2}\big] \\ =&\Pr\big[\mathbf{E}'_{[r-r_2+k-1]|L}(P^j) = \mathbf{E}'_{[r-r_2+k-1]|L}(P^i)\big] \le \frac{r-r_2+k-1}{2^{\mathrm{rank}(A_{12})}} \le \frac{r-1}{2^{\mathrm{rank}(A_{12})}},\end{aligned}$$

for any $j, i$ with $1 \le j < i \le q$. Therefore,

$$\Pr[\text{Bad}] \le \sum_{k=1}^{r_2} C(q, 2) \cdot \frac{r-1}{2^{\mathrm{rank}(A_{12})}} \le \frac{0.5q^2r^2}{2^{\mathrm{rank}(A_{12})}}.$$

When the bad event doesn't occur, the inputs to each round function $f_i$ in $\mathbf{E}_{[r-r_2+1,r]}$ remain collision-free throughout all queries. By Lemma 9, the distinguishing advantage between $\mathbf{E}_{[r-r_2+1,r]}$ and a random permutation is then bounded by $\frac{0.5q^2}{2^{2n}}$. Since the output of $\mathbf{E}_{[r-r_2+1,r]}$ coincides with the output of $\mathbf{E}_{[r]}$, the same bound holds for $\mathbf{E}_{[r]}$ under this condition.

Combining this with the probability of any bad event occurring, the overall distinguishing advantage between $\mathbf{E}_{[r]}$ and a random permutation is at most $\frac{0.5q^2r^2}{2^{\mathrm{rank}(A_{12})}} + \frac{0.5q^2}{2^{2n}}$. □

From Theorem 1 and the attack in Section 6.3, we find that when $T(A_{12}, A_{22}) < \infty$ and $T(A_{12}^\top, A_{11}^\top) < \infty$, UFLM achieves PRP security with minimum rounds $T(A_{12}, A_{22}) + T(A_{12}^\top, A_{11}^\top) + 3$ and resists $\mathcal{O}(\frac{1}{r} \cdot 2^{\frac{\mathrm{rank}(A_{12})}{2}})$ queries. Combined with Corollaries 1 and 2, we obtain that when $\mathrm{rank}(A_{12}) = n$, UFLM achieves PRP security with minimum 3 rounds and the highest query bound $\mathcal{O}(2^{\frac{n}{2}})$. Combined with Corollaries 3 and 4, we obtain that when $\varphi$ satisfies $\varphi^{-1} = \varphi$, UFLM achieves PRP security if and only if $\mathrm{rank}(A_{12}) = n$, with the same number of rounds and query bound as in the general full rank case.

# 7 CCA Security Analysis of UFLM

This section analyzes the CCA security of UFLM when $T(A_{12}, A_{22}) < \infty$ and $T(A_{12}^\top, A_{11}^\top) < \infty$. We first present generic CCA distinguishing attacks against UFLM with at most $2T(A_{12}, A_{22}) + 2$ rounds, and against UFLM with one more round under specific condition. We then prove that UFLM achieves SPRP security at $r = \max\{T(A_{12}, A_{22}) + T(A_{12}^\top, A_{11}^\top) + 3,\ 2T(A_{12}, A_{22}) + 4\}$ rounds.

## 7.1 CCA Attacks for up to $2T(A_{12}, A_{22}) + 2$ Rounds

This section presents a CCA distinguishing attack on UFLM when the number of rounds satisfies $r \le 2T(A_{12}, A_{22}) + 2$, where $T(A_{12}, A_{22}) < \infty$.

Let

$$r_1 = \begin{cases} r - T(A_{12}, A_{22}) - 2, & \text{if } T(A_{12}, A_{22}) + 2 \le r \le 2T(A_{12}, A_{22}) + 2, \\ 0, & \text{if } 1 \le r < T(A_{12}, A_{22}) + 2, \end{cases}$$

and $r_2 = r - r_1 - 1$. We decompose the encryption algorithm of UFLM as $\mathbf{E}_{[r]} = (\mathbf{D}'_{[r_1+2,r]})^{-1} \circ \mathbf{E}_{[r_1+1]} = (\mathbf{D}'_{[r-r_2+1,r]})^{-1} \circ \mathbf{E}_{[r_1+1]}$. Next we study deterministic differential propagations on $\mathbf{E}_{[r_1+1]}$ and on $\mathbf{D}'_{[r-r_2+1,r]}$ separately, as shown in Fig. 8.

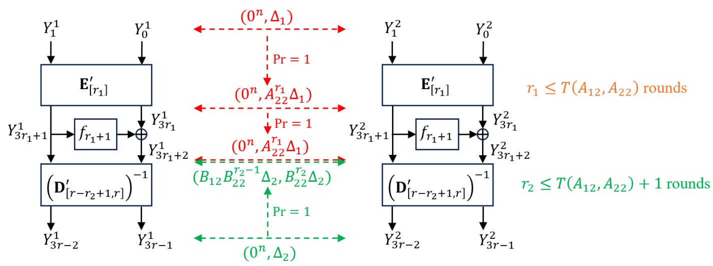



**Fig. 8**: Deterministic differential propagations in $\mathbf{E}_{[r_1+1]} = \mathbf{R}'_{r_1+1} \circ \mathbf{E}'_{[r_1]}$ and $\mathbf{D}'_{[r-r_2+1,r]}$, where the number of rounds satisfies $r_1 \le T(A_{12}, A_{22})$ and $r_2 \le T(A_{12}, A_{22}) + 1$, respectively.

We first study deterministic differential propagation on $\mathbf{E}_{[r_1+1]}$. Decompose it as $\mathbf{R}'_{r_1+1} \circ \mathbf{E}'_{[r_1]}$. The round count of $\mathbf{E}'_{[r_1]}$ satisfies $r_1 \le T(A_{12}, A_{22})$. From Section 4.1, the variant $\mathbf{E}'_{[r_1]}$ possesses the deterministic differential propagation

$$(0^n, \Delta_1) \xrightarrow{\mathbf{E}'_{[r_1]}} (0^n, A_{22}^{r_1}\Delta_1),$$

for some nonzero $\Delta_1 \in \{0,1\}^n$. Furthermore, it holds that

$$(0^n, A_{22}^{r_1}\Delta_1) \xrightarrow{\mathbf{R}'_{r_1+1}} (0^n, A_{22}^{r_1}\Delta_1).$$

Concatenating these two deterministic differential propagations yields

$$(0^n, \Delta_1) \xrightarrow{\mathbf{E}_{[r_1+1]}} (0^n, A_{22}^{r_1}\Delta_1).$$

We then study deterministic differential propagation on $\mathbf{D}'_{[r-r_2+1,r]}$. The round count of it satisfies $r_2 \leq T(A_{12}, A_{22}) + 1$. From Sections 4.1 and 4.2, the process $\mathbf{D}'_{[r-r_2+1,r]}$ possesses the deterministic differential propagation

$$(0^n, \Delta_2) \xrightarrow{\mathbf{D}'_{[r-r_2+1,r]}} (B_{12}B_{22}^{r_2-1}\Delta_2, B_{22}^{r_2}\Delta_2),$$

for some nonzero $\Delta_2 \in \{0,1\}^n$.

Therefore, we can make a successful CCA distinguishing attack against UFLM with $r \leq 2T(A_{12}, A_{22}) + 2$ rounds by applying the boomerang attack [18] described in Section 2.4 with probability close to 1 using only four encryption and decryption queries.

## 7.2 Conditional CCA Attacks for $2T(A_{12}, A_{22}) + 3$ Rounds

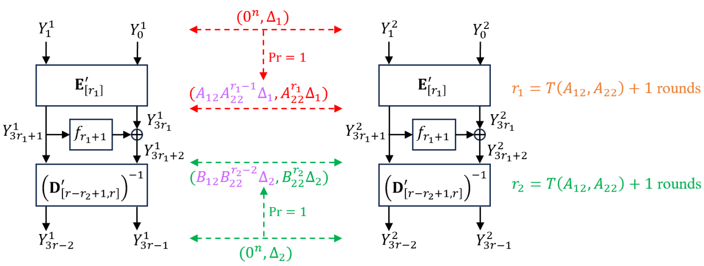



**Fig. 9**: Deterministic differential propagations in $\mathbf{E}'_{[r_1]}$ and $\mathbf{D}'_{[r-r_2+1,r]}$, where the number of rounds satisfies $r_1 = T(A_{12}, A_{22})+1$ and $r_2 = T(A_{12}, A_{22})+1$, respectively.

This section presents a CCA distinguishing attack on UFLM with $r = 2T(A_{12}, A_{22}) + 3$ rounds and $T(A_{12}, A_{22}) < \infty$, which holds under specific condition.

Let $r_1 = r_2 = T(A_{12}, A_{22}) + 1$. We decompose the UFLM encryption algorithm as $\mathbf{E}_{[r]} = (\mathbf{D}'_{[r-r_2+1,r]})^{-1} \circ \mathbf{R}'_{r_1+1} \circ \mathbf{E}'_{[r_1]}$. Next we study deterministic differential propagations on $\mathbf{E}'_{[r_1]}$ and on $\mathbf{D}'_{[r-r_2+1,r]}$ separately, as shown in Fig. 9.

Both $\mathbf{E}'_{[r_1]}$ and $\mathbf{D}'_{[r-r_2+1,r]}$ have $T(A_{12}, A_{22}) + 1$ rounds. From Section 4.2, the transformations $\mathbf{E}'_{[r_1]}$ and $\mathbf{D}'_{[r-r_2+1,r]}$ possess the deterministic differential propagations

$$(0^n, \Delta_1) \xrightarrow{\mathbf{E}'_{[r_1]}} \big(A_{12}A_{22}^{r_1-1}\Delta_1,\ A_{22}^{r_1}\Delta_1\big),$$

$$(0^n, \Delta_2) \xrightarrow{\mathbf{D}'_{[r-r_2+1,r]}} \big(B_{12}B_{22}^{r_2-1}\Delta_2,\ B_{22}^{r_2}\Delta_2\big),$$

for some nonzero $\Delta_1, \Delta_2 \in \{0,1\}^n$, where both $A_{12}A_{22}^{r_1-1}\Delta_1$ and $B_{12}B_{22}^{r_2-1}\Delta_2$ are nonzero.

If the specific condition $A_{12}A_{22}^{r_1-1}\Delta_1 = B_{12}B_{22}^{r_2-1}\Delta_2$ holds. Then the boomerang attack [18] remains effective due to the Feistel switch. Hence, a successful CCA distinguishing attack against UFLM with $r = 2T(A_{12}, A_{22}) + 3$ rounds can be carried out as outlined in Section 2.4, achieving success with probability close to 1 using only four encryption and decryption queries.

*Example 1* When $A_{12}$ is invertible, we have $T(A_{12}, A_{22}) = T(A_{12}^\top, A_{11}^\top) = 0$. Moreover, there exists a pair $(\Delta_1, \Delta_2)$ with nonzero $\Delta_1, \Delta_2$ that satisfies the deterministic differentials

$$(0^n, \Delta_1) \xrightarrow{\mathbf{E}_{[1]}} \big(A_{12}\Delta_1,\ A_{22}\Delta_1\big),$$

$$(0^n, \Delta_2) \xrightarrow{\mathbf{D}'_{[3,3]}} \big(B_{12}\Delta_2,\ B_{22}\Delta_2\big),$$

and fulfills $A_{12}\Delta_1 = B_{12}\Delta_2$. Consequently, the attack described in this section is applicable for 3-round UFLM with an invertible $A_{12}$. In Section C we present a more efficient CCA distinguishing attack for this case, which succeeds with probability almost 1 using only three queries.

### 7.3 CCA Security for $\max\{T(A_{12}, A_{22}) + T(A_{12}^\top, A_{11}^\top) + 3, 2T(A_{12}, A_{22}) + 4\}$ Rounds

This section establishes the SPRP security of UFLM for $r = \max\{T(A_{12}, A_{22}) + T(A_{12}^\top, A_{11}^\top) + 3,\ 2T(A_{12}, A_{22}) + 4\}$ rounds, assuming $T(A_{12}, A_{22}) < \infty$ and $T(A_{12}^\top, A_{11}^\top) < \infty$. The formal statement follows.

**Theorem 2** *Let $\mathbf{E}_{[r]}$ and $\mathbf{D}_{[r]}$ be the encryption and decryption algorithms of $r$-round UFLM with $r = \max\Big\{T(A_{12}, A_{22}) + T(A_{12}^\top, A_{11}^\top) + 3, 2T(A_{12}, A_{22}) + 4\Big\}$, where $T(A_{12}, A_{22}) < \infty, T(A_{12}^\top, A_{11}^\top) < \infty$. Let $\mathcal{A}$ be any sPRP adversary against $r$-round UFLM that makes at most $q$ queries. Then*

$$\mathrm{Adv}^{\mathrm{sprp}}_{\mathbf{E}_{[r]}}(\mathcal{A}) \leq \frac{0.5q^2r^2}{2^{\mathrm{rank}(A_{12})}} + \frac{q^2}{2^{2n}}.$$

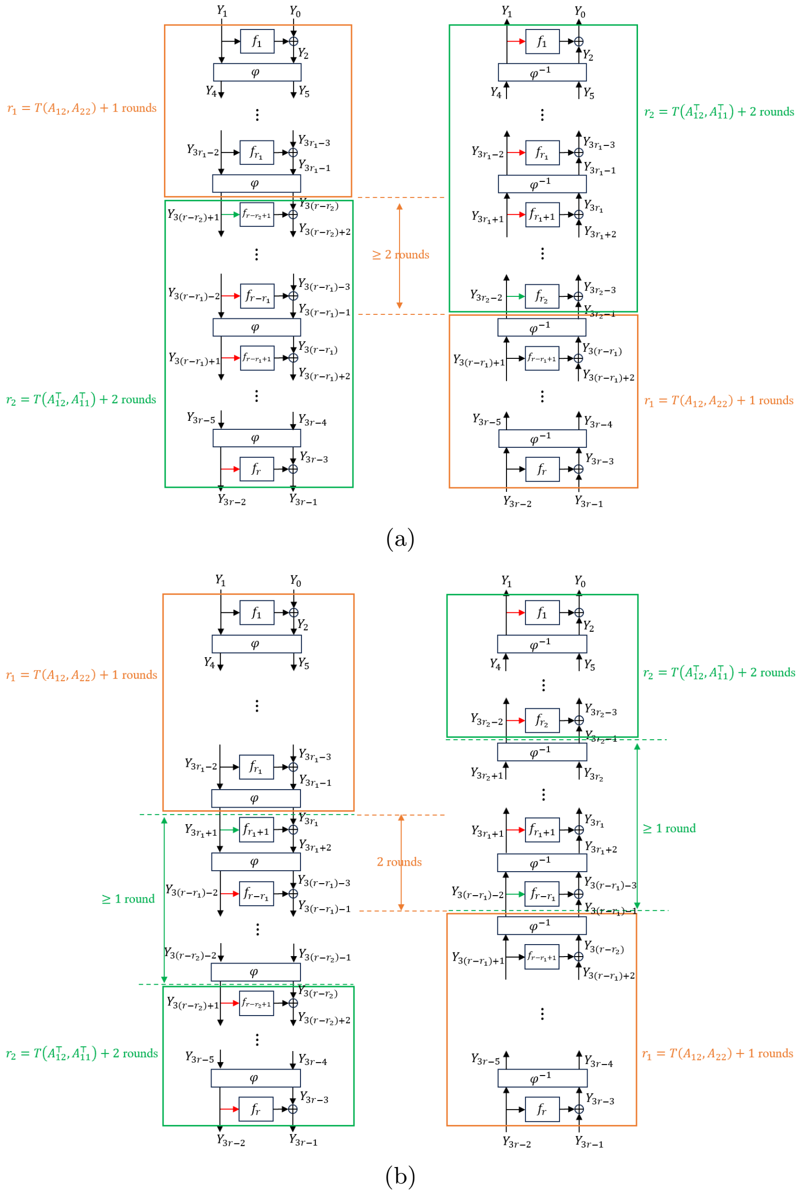



**Fig. 10**: Illustration of the CCA proof for UFLM when the number of rounds satisfies $r = \max\{T(A_{12}, A_{22}) + T(A_{12}^\top, A_{11}^\top) + 3, 2T(A_{12}, A_{22}) + 4\}$. Branches in green and red correspond to low collision probability and low differential probability, respectively. Let $r_1 := T(A_{12}, A_{22}) + 1$ and $r_2 := T(A_{12}^\top, A_{11}^\top) + 2$. (a) The case when $r_2 \geq r_1 + 2$ and then $r = r_1 + r_2$. (b) The case when $r_2 \leq r_1 + 1$ and then $r = 2r_1 + 2$.

*Proof* Let $r_1 \coloneqq T(A_{12}, A_{22}) + 1$ and $r_2 \coloneqq T(A_{12}^\top, A_{11}^\top) + 2$. Then

$$r = \begin{cases} r_1 + r_2, & \text{if } r_2 \geq r_1 + 2, \\ 2r_1 + 2, & \text{if } r_2 \leq r_1 + 1 \end{cases} \tag{17}$$

The proof proceeds in two steps. First, we bound the collision probability between the inputs to any round function $f_i$ in $\mathbf{E}_{[r-r_2+1,r]}$ within each encryption query and inputs to that same round function from any prior encryption or decryption query; we also bound collisions for the inputs to any round function $f_i$ in $\mathbf{D}_{[r_2]}$ within each decryption query against all previous encryption or decryption queries. Second, conditioned on the absence of these collisions, both $\mathbf{E}_{[r-r_2+1,r]}$ and $\mathbf{D}_{[r_2]}$ are PRPs, hence $\mathbf{E}_{[r]}$ and $\mathbf{D}_{[r]}$ are PRPs. This argument is illustrated in Figure 10.

We assume that all encryption queries use distinct plaintexts and all decryption queries use distinct ciphertexts. Also, no decryption query repeats a ciphertext obtained from an earlier encryption query, and no encryption query repeats a plaintext obtained from an earlier decryption query. For $q$ queries to UFLM, we denote by $P^i$ the plaintext, $C^i$ the ciphertext and $Y_j^i$ the value of the variable $Y_j$ in the $i$-th query. Let $\mathcal{Q}_e$ and $\mathcal{Q}_d$ denote the index sets for encryption and decryption queries, respectively, with $|\mathcal{Q}_e| + |\mathcal{Q}_d| = q$.

First, we bound the collision probability. To capture these incidents, we define 4 bad events. For $i \in \mathcal{Q}_e$,

- $\mathrm{Bad}_1$: $Y^j_{3(r-r_2+k)-2} = Y^i_{3(r-r_2+k)-2}$ for some $1 \leq j < i \leq q, j \in \mathcal{Q}_e, k \in \{1, 2, \ldots, r_2\}$.
- $\mathrm{Bad}_2$: $Y^j_{3(r-r_2+k)-2} = Y^i_{3(r-r_2+k)-2}$ for some $1 \leq j < i \leq q, j \in \mathcal{Q}_d, k \in \{1, 2, \ldots, r_2\}$.

For $i \in \mathcal{Q}_d$,

- $\mathrm{Bad}_3$: $Y^j_{3k-2} = Y^i_{3k-2}$ for some $1 \leq j < i \leq q, j \in \mathcal{Q}_d, k \in \{1, 2, \ldots, r_2\}$.
- $\mathrm{Bad}_4$: $Y^j_{3k-2} = Y^i_{3k-2}$ for some $1 \leq j < i \leq q, j \in \mathcal{Q}_e, k \in \{1, 2, \ldots, r_2\}$.

First, we bound the probability of event $\mathrm{Bad}_1$. For every $k \in \{1, 2, \ldots, r_2\}$, the number of rounds $r - r_2 + k - 1 \geq r_1$. Then for any $j, i$ with $1 \leq j < i \leq q$ and $j \in \mathcal{Q}_e, i \in \mathcal{Q}_e$,

$$\begin{aligned} &\Pr\big[Y^j_{3(r-r_2+k)-2} = Y^i_{3(r-r_2+k)-2}\big] \\ &= \Pr\big[\mathbf{E}'_{[r-r_2+k-1]|L}(P^j) = \mathbf{E}'_{[r-r_2+k-1]|L}(P^i)\big] \leq \frac{r - r_2 + k - 1}{2^{\mathrm{rank}(A_{12})}} \leq \frac{r-1}{2^{\mathrm{rank}(A_{12})}} \end{aligned}$$

by Lemma 5. Therefore,

$$\Pr[\mathrm{Bad}_1] \leq \sum_{k=1}^{r_2} \sum_{i \in \mathcal{Q}_e} \sum_{j \in \mathcal{Q}_e, j<i} \frac{r-1}{2^{\mathrm{rank}(A_{12})}}.$$

Next, we bound the probability of event $\mathrm{Bad}_2$. For any pair $j, i$ with $1 \leq j < i \leq q$ and $j \in \mathcal{Q}_d, i \in \mathcal{Q}_e$,

$$\begin{aligned} &\Pr\big[Y^j_{3(r-r_2+k)-2} = Y^i_{3(r-r_2+k)-2}\big] \\ &= \Pr[Y^j_{3(r-r_2+k)-2} \oplus Y^c_{3(r-r_2+k)-2} = Y^c_{3(r-r_2+k)-2} \oplus Y^i_{3(r-r_2+k)-2}] \\ &= \Pr\begin{bmatrix} \mathbf{D}'_{[r-r_2+k+1,r]|L}(C^j) \oplus \mathbf{D}'_{[r-r_2+k+1,r]|L}(C^c) \\ = \mathbf{E}'_{[r-r_2+k-1]|L}(P^c) \oplus \mathbf{E}'_{[r-r_2+k-1]|L}(P^i) \end{bmatrix} \end{aligned} \tag{18}$$

where $c \in \{1, \ldots, i-1\} \setminus \{j\}$ if $\{1, \ldots, i-1\} \setminus \{j\} \neq \emptyset$; else, $c = j$. Let

$$\begin{aligned} \nabla_{j,c} &\coloneqq \mathbf{D}'_{[r-r_2+k+1,r]|L}(C^j) \oplus \mathbf{D}'_{[r-r_2+k+1,r]|L}(C^c), \\ \Delta_{c,i} &\coloneqq \mathbf{E}'_{[r-r_2+k-1]|L}(P^c) \oplus \mathbf{E}'_{[r-r_2+k-1]|L}(P^i). \end{aligned}$$

(1) If $\{1,\ldots,i-1\}\setminus\{j\}\neq\emptyset$ and $c\in\{1,\ldots,i-1\}\setminus\{j\}$,

- When $r_2\geq r_1+2$ and $k=1$, the number of rounds of $\mathbf{D}'_{[r-r_2+k+1,r]}$ is $r_2-1\geq r_1+1$ based on Eq. (17). Then by Lemma 8,

$$\Pr[\mathbf{D}'_{[r-r_2+k+1,r]|L}(C^j)\oplus\mathbf{D}'_{[r-r_2+k+1,r]|L}(C^c)=\Delta_{c,i}]\leq\frac{r_2-1}{2^{\mathrm{rank}(A_{12})}}\leq\frac{r-1}{2^{\mathrm{rank}(A_{12})}}$$

for any $c\in\{1,\ldots,i-1\}\setminus\{j\}$.

- Else when $r_2\geq r_1+2$ and $k\in\{2,\ldots,r_2\}$ or $r_2\leq r_1+1$ and $k\in\{1,2,\ldots,r_2\}$ , the number of rounds of $\mathbf{E}'_{[r-r_2+k-1]|L}$ is $r-r_2+k-1\geq r_1+1$ based on Eq. (17). Then by Lemma 7,

$$\Pr[\mathbf{E}'_{[r-r_2+k-1]|L}(P^c)\oplus\mathbf{E}'_{[r-r_2+k-1]|L}(P^i)=\nabla_{j,c}]\leq\frac{r-r_2+k-1}{2^{\mathrm{rank}(A_{12})}}\leq\frac{r-1}{2^{\mathrm{rank}(A_{12})}}$$

for any $c\in\{1,\ldots,i-1\}\setminus\{j\}$.

Thus Eq. (18) is bound by $\frac{r-1}{2^{\mathrm{rank}(A_{12})}}$.

(2) Else if $\{1,\ldots,i-1\}\setminus\{j\}=\emptyset$ and $c=j$,

$$Eq.\ (18)=\Pr[\mathbf{E}'_{[r-r_2+k-1]|L}(P^j)\oplus\mathbf{E}'_{[r-r_2+k-1]|L}(P^i)=0^n]. \tag{19}$$

The number of rounds of $\mathbf{E}'_{[r-r_2+k-1]|L}$ is $r-r_2+k-1\geq r_1$ based on Eq. (17). Then by Lemma 5,

$$Eq.\ (19)\leq\frac{r-r_2+k-1}{2^{\mathrm{rank}(A_{12})}}\leq\frac{r-1}{2^{\mathrm{rank}(A_{12})}}.$$

Consequently,

$$\Pr[\mathrm{Bad}_2]\leq\sum_{k=1}^{r_2}\sum_{i\in\mathcal{Q}_e}\sum_{j\in\mathcal{Q}_d,j<i}\frac{r-1}{2^{\mathrm{rank}(A_{12})}}.$$

Due to the perfect symmetry between the UFLM structures underlying decryption queries and encryption queries, the probabilities of events $\mathrm{Bad}_3$ and $\mathrm{Bad}_4$ are bounded by an analogous analysis.

$$\Pr[\mathrm{Bad}_3]\leq\sum_{k=1}^{r_2}\sum_{i\in\mathcal{Q}_d}\sum_{j\in\mathcal{Q}_d,j<i}\frac{r-1}{2^{\mathrm{rank}(A_{12})}},$$

$$\Pr[\mathrm{Bad}_4]\leq\sum_{k=1}^{r_2}\sum_{i\in\mathcal{Q}_d}\sum_{j\in\mathcal{Q}_e,j<i}\frac{r-1}{2^{\mathrm{rank}(A_{12})}}.$$

Therefore,

$$\sum_{\ell=1}^{4}\Pr[\mathrm{Bad}_\ell]\leq\sum_{k=1}^{r_2}C(q,2)\cdot\frac{r-1}{2^{\mathrm{rank}(A_{12})}}\leq\frac{0.5q^2r^2}{2^{\mathrm{rank}(A_{12})}}.$$

When none of these bad events occur, the inputs to the final $r_2$ round functions $f_{r-r_2+1},\ldots,f_r$ remain collision-free across all encryption queries, and similarly, the inputs to the $r_2$ round functions $f_1,\ldots,f_{r_2}$ remain collision-free across all decryption queries. By Lemmas 9 and 10, the distinguishing advantages between $\mathbf{E}_{[r-r_2+1,r]}$ (resp. $\mathbf{D}_{[r_2]}$) and a uniformly random permutation are bounded by $\frac{0.5q^2}{2^{2n}}$. Because the output of $\mathbf{E}_{[r-r_2+1,r]}$ coincides with that of $\mathbf{E}_{[r]}$, and the output of $\mathbf{D}_{[r_2]}$ coincides with that of $\mathbf{D}_{[r]}$, the same bound applies to $\mathbf{E}_{[r]}$ and $\mathbf{D}_{[r]}$ in this setting.

Taking into account the probabilities of all bad events, the overall distinguishing advantage between the $(\mathbf{E}_{[r]},\mathbf{D}_{[r]})$ and $(\pi,\pi^{-1})$ for $\pi\leftarrow\!\!\$\ \mathsf{Perm}(n)$ satisfies $\frac{0.5q^2r^2}{2^{\mathrm{rank}(A_{12})}}+\frac{q^2}{2^{2n}}$. $\square$

From Theorem 2 and the attacks in Sections 6.3, 7.1 and 7.2, we know that when $T(A_{12}, A_{22}) < \infty$ and $T(A_{12}^\top, A_{11}^\top) < \infty$, UFLM achieves SPRP security with minimum rounds $\max\{T(A_{12}, A_{22}) + T(A_{12}^\top, A_{11}^\top) + 3, 2T(A_{12}, A_{22}) + 4\}$ and resists $\mathcal{O}(\frac{1}{r} \cdot 2^{\frac{\mathrm{rank}(A_{12})}{2}})$ queries. Combined with Corollaries 1 and 2, we find that when $\mathrm{rank}(A_{12}) = n$, the minimum number of rounds for SPRP security is 4, and the query bound reaches its maximum $\mathcal{O}(2^{\frac{n}{2}})$. Combined with Corollaries 3 and 4, we obtain that when $\varphi$ satisfies $\varphi^{-1} = \varphi$, UFLM achieves SPRP security if and only if $\mathrm{rank}(A_{12}) = n$, with the same round number and query bound as in the general full rank case.

# 8 Conclusion

This paper provides a systematic and comprehensive security evaluation of the UFLM framework, specifically focusing on its indistinguishability from an ideal random permutation. Our findings reveal distinct security landscapes for UFLM based on the properties of its crucial linear mapping $\varphi$, particularly the invertibility of its $A_{12}$ submatrix.

For UFLM where $A_{12}$ is invertible, we establish that 2-round UFLM under CPA and 3-round UFLM under CCA are vulnerable to distinguishing attacks with a minimal number of queries. Conversely, 3-round UFLM under CPA and 4-round UFLM under CCA achieve PRP and sPRP security, respectively, with query complexities proportional to $\mathcal{O}(2^{n/2})$. These results confirm and extend known security results of its well-known instances like Feistel and Lai-Massey.

Critically, our work highlights a significant vulnerability when the $A_{12}$ matrix is non-invertible. This non-invertibility gives rise to deterministic differential propagations that drastically reduce the security of UFLM. We demonstrate that even for higher round numbers, the presence of such non-invertible components allows for distinguishing attacks with very few queries. Importantly, the exact round threshold that separates secure from insecure instances is jointly determined by the matrices $A_{11}, A_{12}$, and $A_{22}$. Despite these weaknesses, we also establish new PRP and sPRP security bounds for UFLM in this setting, showing that sufficient rounds can restore security, albeit with complexities that still depend on the rank of $A_{12}$.

Our work further establishes that when an involutory matrix serves as the linear layer, the sub-matrix $A_{12}$ must be of full rank; otherwise, the construction cannot achieve pseudorandom security for any number of rounds.

In summary, this research provides a deeper, more nuanced understanding of the UFLM framework's security, revealing how fundamental algebraic properties of its linear transformation directly dictate its resilience against distinguishing attacks. This work emphasizes that the design of the linear layer is paramount for achieving robust security in iterative block ciphers. The results serve as a crucial guide for designers of cryptographic primitives utilizing the UFLM architecture.

**Acknowledgements.** We thank Zhengyi Dai for answering questions about the UFLM structure in his paper. This work was supported by the Research Startup Fund of Ningbo University of Technology (25KQ076) and by the National Key Laboratory of Security Communication Foundation (2024, 6142103042409).

## Declarations

- The authors have no relevant financial or non-financial interests to disclose.
- The authors have no competing interests to declare that are relevant to the content of this article.
- All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.
- The authors have no financial or proprietary interests in any material discussed in this article.

# Appendix A Proof of Lemma 2

To prove Lemma 2, we first need the following lemma.

*Lemma 11* For any $2n \times 2n$ invertible matrix over $GF(2)$ and its inversion defined as

$$\varphi = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}, \varphi^{-1} = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}, \tag{A1}$$

where $A_{vw}, B_{vw}$ for $v, w \in \{1, 2\}$ are $n \times n$ matrices, for any $i \in \{0, 1, 2, 3, \ldots\}$ assume that events $\mathsf{A}_i$ and $\mathsf{B}_i$ happen, then it holds that

$$\begin{cases} \overline{\mathsf{A}_{i+1}} \Leftrightarrow \overline{\mathsf{B}_{i+1}}, & \text{(A2)} \\ \mathsf{A}_{i+1} \Leftrightarrow \mathsf{B}_{i+1}, & \text{(A3)} \end{cases}$$

where

$$\text{event } \mathsf{A}_i := \left( \exists \Delta \in \{0,1\}^n \setminus \{0^n\} \text{ such that} \begin{cases} A_{12}\Delta = 0^n, \\ A_{12}A_{22}\Delta = 0^n, \\ \vdots \\ A_{12}A_{22}^i\Delta = 0^n \end{cases} \right). \tag{A4}$$

$$\text{event } \mathsf{B}_i := \left( \exists \Delta \in \{0,1\}^n \setminus \{0^n\} \text{ such that} \begin{cases} B_{12}\Delta = 0^n, \\ B_{12}B_{22}\Delta = 0^n, \\ \vdots \\ B_{12}B_{22}^i\Delta = 0^n \end{cases} \right). \tag{A5}$$

*Proof* For the reason that Eq. (A2) implies Eq. (A3), so we only prove Eq. (A2).

Firstly, we prove

$$\overline{\mathsf{A}_{i+1}} \Rightarrow \overline{\mathsf{B}_{i+1}}. \tag{A6}$$

This implies that any $\Delta \in \{0,1\}^n \setminus \{0^n\}$ which satisfies

$$\begin{cases} B_{12}\Delta = 0^n, & \text{(A7)} \\ B_{12}B_{22}\Delta = 0^n, & \text{(A8)} \\ \vdots & \text{(A9)} \\ B_{12}B_{22}^i\Delta = 0^n, & \text{(A10)} \end{cases}$$

we need to prove $B_{12}B_{22}^{i+1}\Delta \neq 0^n$.

For $j = 0, 1, \ldots, i$, we have

$$\left.\begin{aligned} Eq.\ (5) \Rightarrow A_{11}B_{12}B_{22}^{j}\Delta \oplus A_{12}B_{22}^{j+1}\Delta = 0^n \\ B_{12}B_{22}^{j}\Delta = 0^n \end{aligned}\right\} \Rightarrow A_{12}B_{22}^{j+1}\Delta = 0^n, \tag{A11}$$

$$\left.\begin{aligned} Eq.\ (6) \Rightarrow A_{21}B_{12}B_{22}^{j}\Delta \oplus A_{22}B_{22}^{j+1}\Delta = B_{22}^{j}\Delta \\ B_{12}B_{22}^{j}\Delta = 0^n \end{aligned}\right\} \Rightarrow A_{22}B_{22}^{j+1}\Delta = B_{22}^{j}\Delta. \tag{A12}$$

The invertibility of $\varphi^{-1}$ implies the rank of $\begin{pmatrix} B_{12} \\ B_{22} \end{pmatrix}$ is $n$. This means that there does not exist a non-zero $y$ such that $\begin{pmatrix} B_{12} \\ B_{22} \end{pmatrix} y = 0^n$. Then from $B_{12}(B_{22}^{i}\Delta) = 0^n$, we have $B_{22}^{i+1}\Delta = B_{22}(B_{22}^{i}\Delta) \neq 0^n$. From Eqs. (A11) and (A12), it follows that

$$\begin{cases} A_{12}(B_{22}^{i+1}\Delta) = 0^n, \\ A_{12}A_{22}(B_{22}^{i+1}\Delta) = A_{12}B_{22}^{i}\Delta = 0^n, \\ A_{12}A_{22}^{2}(B_{22}^{i+1}\Delta) = A_{12}A_{22}B_{22}^{i}\Delta = A_{12}B_{22}^{i-1}\Delta = 0^n, \\ \qquad \vdots \\ A_{12}A_{22}^{i}(B_{22}^{i+1}\Delta) = A_{12}A_{22}^{i-1}B_{22}^{i}\Delta = \ldots = A_{12}B_{22}\Delta = 0^n, \end{cases}$$

Combining the above equations with event $\overline{\mathsf{A}_{i+1}}$, we have $A_{12}A_{22}^{i+1}(B_{22}^{i+1}\Delta) \neq 0^n$. In fact,

$$A_{12}A_{22}^{i+1}(B_{22}^{i+1}\Delta) = A_{12}A_{22}^{i}B_{22}^{i}\Delta = \ldots = A_{12}\Delta.$$

Therefore, $A_{12}\Delta \neq 0^n$.

From the above proof, we conclude that for any $\Delta \in \{0,1\}^n \setminus \{0^n\}$,

$$\left.\begin{aligned} &Eq.\ (A7) \\ &Eq.\ (A8) \\ &\quad \vdots \\ &Eq.\ (A10) \end{aligned}\right\} \Rightarrow A_{12}\Delta \neq 0^n.$$

Now, we have known $A_{12}(B_{22}\Delta) = 0^n$ for $B_{22}\Delta \neq 0^n$, where $B_{22}\Delta \neq 0^n$ is from the rank of $\begin{pmatrix} B_{12} \\ B_{22} \end{pmatrix}$ is $n$ and $B_{12}\Delta = 0^n$. Therefore,

$$\begin{cases} \quad\ B_{12}(B_{22}\Delta) \neq 0^n, \\ \text{or } B_{12}B_{22}(B_{22}\Delta) \neq 0^n, \\ \qquad \vdots \\ \text{or } B_{12}B_{22}^{i}(B_{22}\Delta) \neq 0^n \end{cases}$$

From Eqs. (A8) to (A10), we have already known that $B_{12}B_{22}^{j}(B_{22}\Delta) = 0^n$ for $j = 0, 1, 2, \ldots, i-1$. Then, $B_{12}B_{22}^{i+1}\Delta = B_{12}B_{22}^{i}(B_{22}\Delta) \neq 0^n$ must hold. So we conclude Eq. (A6).

On the other hand, we can also prove

$$\overline{\mathsf{A}_{i+1}} \Leftarrow \overline{\mathsf{B}_{i+1}},$$

by interchanging the positions of $A_{ij}$ and $B_{ij}$ in the above proof.

Then we conclude the proof. □

Then we can prove Lemma 2. For $T(A_{12}, A_{22}) = i$ where $i \geq 1$, the event $\mathsf{A}_{i-1} \wedge \overline{\mathsf{A}_i}$ occurs. From Lemma 1, we have $[\mathsf{A}_0 \Leftrightarrow \mathsf{B}_0]$. If $[\mathsf{A}_j \Leftrightarrow \mathsf{B}_j]$, we have $[\mathsf{A}_{j+1} \Leftrightarrow \mathsf{B}_{j+1}]$ from Lemma 11. Therefore, based on mathematical induction, we obtain $[\mathsf{A}_{i-1} \Leftrightarrow \mathsf{B}_{i-1}]$. Based on Lemma 11, it follows that $\overline{\mathsf{A}_i} \Leftrightarrow \overline{\mathsf{B}_i}$. Therefore, event $\mathsf{B}_{i-1} \wedge \overline{\mathsf{B}_i}$ occurs, which implies that $T(B_{12}, B_{22}) = i$. So $T(A_{12}, A_{22}) = T(B_{12}, B_{22})$ for $i \geq 1$ as well.

## Appendix B Proof of Lemma 4

From $\varphi = \varphi^{-1}$ we have $\varphi^2 = I$. Expanding $\varphi^2 = I$ gives

$$A_{11}A_{12} \oplus A_{12}A_{22} = \mathbf{0} \quad \Longrightarrow \quad A_{11}A_{12} = A_{12}A_{22}.$$

By induction we obtain the general term $A_{11}^i A_{12} = A_{12}A_{22}^i$ also holds for every non-negative integer $i$ by

$$A_{11}^i A_{12} = A_{11}^{i-1}(A_{11}A_{12}) = A_{11}^{i-1}A_{12}A_{22} = \cdots = A_{12}A_{22}^i.$$

Consequently,

$$V_{A_{12}^\top, A_{11}^\top, i} = \begin{pmatrix} A_{12}^\top \\ A_{12}^\top A_{11}^\top \\ \vdots \\ A_{12}^\top \left(A_{11}^\top\right)^i \end{pmatrix} = \begin{pmatrix} A_{12} & A_{11}A_{12} & \ldots & A_{11}^i A_{12} \end{pmatrix}^\top = \begin{pmatrix} A_{12} & A_{12}A_{22} & \ldots & A_{12}A_{22}^i \end{pmatrix}^\top,$$

and

$$V_{A_{12}, A_{22}, i} = \begin{pmatrix} A_{12} \\ A_{12}A_{22} \\ \vdots \\ A_{12}A_{22}^i \end{pmatrix} = \begin{pmatrix} A_{12} \\ A_{11}A_{12} \\ \vdots \\ A_{11}^i A_{12} \end{pmatrix}.$$

Now,

$$\operatorname{rank}\left(V_{A_{12}^\top, A_{11}^\top, i}\right) = \operatorname{rank}\left(V_{A_{12}^\top, A_{11}^\top, i}^\top\right) = \operatorname{rank}\begin{pmatrix} A_{12} & A_{12}A_{22} & \ldots & A_{12}A_{22}^i \end{pmatrix},$$

where each block has the form $A_{12}A_{22}^k$. Because $\operatorname{Col}(XY) \subseteq \operatorname{Col}(X)$ for any matrices $X, Y$, every column of $A_{12}A_{22}^k$ lies in $\operatorname{Col}(A_{12})$. Hence

$$\operatorname{rank}\left(V_{A_{12}^\top, A_{11}^\top, i}\right) = \operatorname{rank}(A_{12}).$$

Similarly, for $V_{A_{12}, A_{22}, i}$ every block is of the form $A_{11}^k A_{12}$. Using $\operatorname{Row}(YX) \subseteq \operatorname{Row}(X)$, we obtain

$$\operatorname{rank}\left(V_{A_{12}, A_{22}, i}\right) = \operatorname{rowrank}\left(V_{A_{12}, A_{22}, i}\right) = \operatorname{rowrank}(A_{12}) = \operatorname{rank}(A_{12}).$$

Thus

$$\operatorname{rank}\left(V_{A_{12}, A_{22}, i}\right) = \operatorname{rank}\left(V_{A_{12}^\top, A_{11}^\top, i}\right) = \operatorname{rank}(A_{12}).$$

for every non-negative integer $i$.

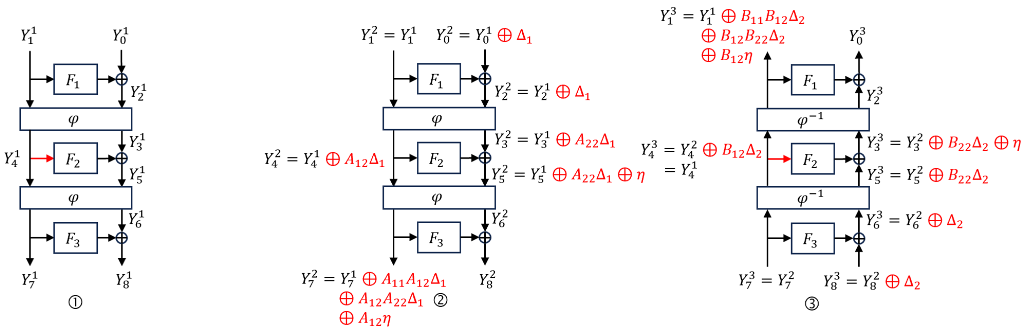



**Fig. C1**: The process of three queries to $\mathbf{E}_{[3]}, \mathbf{E}_{[3]}$, and $\mathbf{D}_{[3]}$. The values in the red branches are identical.

# Appendix C CCA Attack on 3-round UFLM with invertibe $A_{12}$

We provide successful CCA distinguishing attack against 3-round UFLM with invertible $A_{12}$ as follows.

(1) Choose an arbitrary element $\Delta_1, \Delta_2 \in \{0,1\}^n \setminus \{0^n\}$ such that $B_{12}\Delta_2 = A_{12}\Delta_1$.
(2) Query the encryption oracle on plaintext $(Y_1^1, Y_0^1)$ to get $(Y_7^1, Y_8^1)$.
(3) Query the decryption oracle on ciphertext $(Y_7^3, Y_8^3) = (Y_7^2, Y_8^2 \oplus \Delta_2)$ to get $(Y_1^3, Y_0^3)$.
(4) If

$$\begin{aligned} &A_{12}^{-1}((Y_7^2 \oplus Y_7^1) \oplus A_{11}A_{12}\Delta_1 \oplus A_{12}A_{22}\Delta_1) \\ =&B_{12}^{-1}((Y_1^3 \oplus Y_1^2) \oplus B_{11}B_{12}\Delta_2 \oplus B_{12}B_{22}\Delta_2), \end{aligned} \tag{C13}$$

output 1; else, output 0.

For $\mathbf{E}_{[3]}$ and $\mathbf{D}_{[3]}$, the three queries are illustrated in Fig. C1. For the second query, from $(Y_1^2, Y_0^2) = (Y_1^1, Y_0^1 \oplus \Delta_1)$, it follows that

$$Y_7^2 = Y_7^1 \oplus A_{11}A_{12}\Delta_1 \oplus A_{12}A_{22}\Delta_1 \oplus A_{12}\eta \tag{C14}$$

where $\eta = F_2(Y_4^1) \oplus F_2(Y_4^1 \oplus A_{12}\Delta_1)$. For the third query, it follows that

$$Y_1^3 = Y_1^2 \oplus B_{11}B_{12}\Delta_2 \oplus B_{12}B_{22}\Delta_2 \oplus B_{12}\eta. \tag{C15}$$

Eqs. (C14) and (C15) lead to Eq. (C13). Therefore, $\mathcal{A}$ outputs 1 with probability 1. However, for $\pi$ and $\pi^{-1}$, by their randomness, $\mathcal{A}$ outputs 1 with probability at most $\frac{1}{2^{n-2}}$. Therefore, the advantage of $\mathcal{A}$ in distinguishing $(\mathbf{E}_{[3]}, \mathbf{D}_{[3]})$ from $(\pi, \pi^{-1})$ is almost 1 using 3 encryption and decryption queries.]

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